

Applications of Vector Algebra

Ex 6.1

Question 1.

Prove by vector method that if a line is drawn from the centre of a circle to the midpoint of a chord, then the line is perpendicular to the chord.

Solution:

Let 'C' be the mid point of the chord AB. Take 'O' as the centre of the circle.

Since, $OA = OB$ (Radii)

Let $\vec{OA} = \vec{a}$; $\vec{OB} = \vec{b}$; $\vec{OC} = \frac{\vec{a} + \vec{b}}{2}$

To prove: $\vec{OC} \perp \vec{AB}$

$$\vec{OC} \cdot \vec{AB} = (\frac{\vec{a} + \vec{b}}{2}) \cdot (\vec{OB} - \vec{OA}) = (\frac{\vec{a} + \vec{b}}{2}) \cdot (\vec{b} - \vec{a}) = \frac{1}{2}[(\vec{b} \cdot \vec{b}) - (\vec{a} \cdot \vec{a})] = \frac{1}{2}(|\vec{b}|^2 - |\vec{a}|^2) = \frac{1}{2}(OB^2 - OA^2) = 0$$

$\therefore \vec{OC} \perp \vec{AB}$. Hence the result.

Question 2.

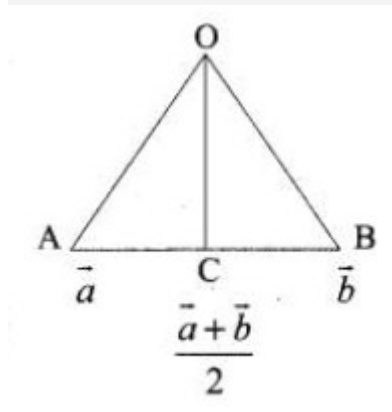
Prove by vector method that the median to the base of an isosceles triangle is perpendicular to the base.

Solution:

Let OAB be an isosceles triangle with $OA = OB$

Let OC be the median to the base AB

C is the midpoint of AB



Take O as origin.

- Let $\vec{OA} = \vec{a}$; $\vec{OB} = \vec{b}$; $\vec{OC} = \frac{\vec{a} + \vec{b}}{2}$

To prove $\vec{OC} \cdot \vec{AB} = 0$

$$\vec{OC} \cdot \vec{AB} = (\frac{\vec{a} + \vec{b}}{2}) \cdot (\vec{OB} - \vec{OA}) = (\frac{\vec{a} + \vec{b}}{2}) \cdot (\vec{b} - \vec{a}) = \frac{1}{2}[(\vec{b} \cdot \vec{b}) - (\vec{a} \cdot \vec{a})] = \frac{1}{2}(|\vec{b}|^2 - |\vec{a}|^2) = \frac{1}{2}(OB^2 - OA^2) = 0$$

$\therefore \vec{OC} \perp \vec{AB}$. Hence the result.

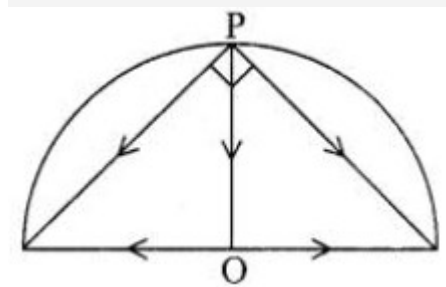
Question 3.

Prove by vector method that an angle in a semi-circle is a right angle.

Solution:

Let AB be the diameter of the circle with centre 'O'

Let P be any point on the semi-circle.



- To prove $\angle APB = 90^\circ$

We have $OA = OB = OP$ (radii)

Now

$$\begin{aligned} \vec{PA} &= \vec{PO} + \vec{OA} \quad \vec{PB} = \vec{PO} + \vec{OB} = \vec{PO} - \vec{OA} \quad (\text{since } \vec{OB} = -\vec{OA}) \\ \therefore \vec{PA} \cdot \vec{PB} &= (\vec{PO} + \vec{OA}) \cdot (\vec{PO} - \vec{OA}) = (\vec{PO})^2 - (\vec{OA})^2 = (\vec{PO})^2 - (\vec{OA})^2 \therefore \vec{PA} \cdot \vec{PB} = 0 \end{aligned}$$

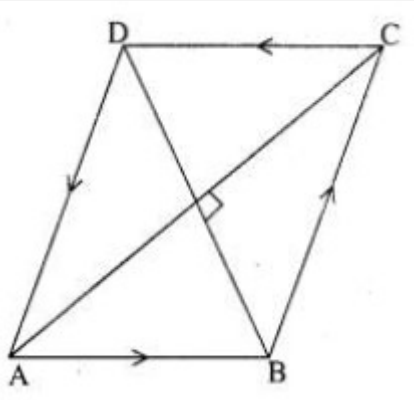
This gives $\angle APB = 90^\circ$. Hence the result.

Question 4.

Prove by vector method that the diagonals of a rhombus bisect each other at right angles.

Solution:

Let ABCD be a rhombus



To prove $\vec{AC} \cdot \vec{BD} = 0$

We have $AB = BC = CD = DA$

Now

$$\begin{aligned} \vec{AC} &= \vec{AB} + \vec{BC} \quad \vec{BD} = \vec{BC} + \vec{CD} = \vec{BC} - \vec{AB} \quad (\text{since } \vec{CD} = -\vec{AB}) \\ \therefore \vec{BD} &= (\vec{BC} - \vec{AB}) \cdot (\vec{BC} - \vec{AB}) = (\vec{BC})^2 - (\vec{AB})^2 = (\vec{BC})^2 - (\vec{AB})^2 \therefore \vec{AC} \cdot \vec{BD} = 0 \end{aligned}$$

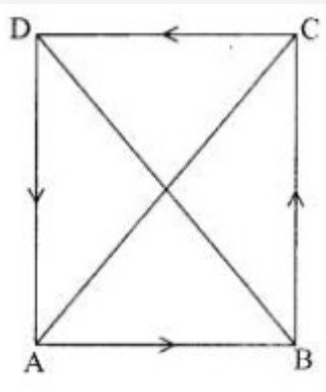
$\therefore \vec{AC} \perp \vec{BD}$. Hence the result.

Question 5.

Using vector method, prove that if the diagonals of a parallelogram are equal, then it is a rectangle

Solution:

Let ABCD be a parallelogram To prove ABCD be a rectangle provided the diagonals are equal.



But

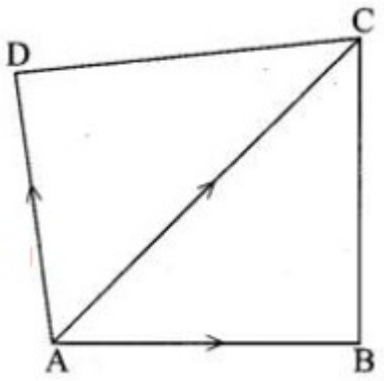
$$\begin{aligned} \vec{AC} &= \vec{AB} + \vec{BC} \quad \vec{BD} = \vec{BC} + \vec{CD} = \vec{BC} - \vec{AB} \\ (\vec{AB} + \vec{BC})^2 &= (\vec{BC} - \vec{AB})^2 \quad (\vec{AB})^2 + (\vec{BC})^2 + 2\vec{AB} \cdot \vec{BC} = (\vec{BC})^2 + (\vec{AB})^2 - 2\vec{AB} \cdot \vec{BC} \\ 4\vec{AB} \cdot \vec{BC} &= 0 \quad \vec{AB} \cdot \vec{BC} = 0 \quad \vec{AB} \perp \vec{BC} \Rightarrow \text{ABCD is a rectangle.} \end{aligned}$$

Question 6.

Prove by vector method that the area of the quadrilateral ABCD having diagonals AC and BD is $\frac{1}{2} |\vec{AC} \times \vec{BD}|$

Solution:

Vector area of quadrilateral ABCD = { Vector area of $\triangle ABC$ } + { Vector area of $\triangle ACD$ }



$$=12(\vec{AB} \times \vec{AC}) + 12(\vec{AC} \times \vec{AD}) = -12(\vec{AC} \times \vec{AB}) + 12(\vec{AC} \times \vec{AD}) = 12\vec{AC} \times [\vec{AB} + \vec{AD}] = 12\vec{AC} \times \vec{BD}$$

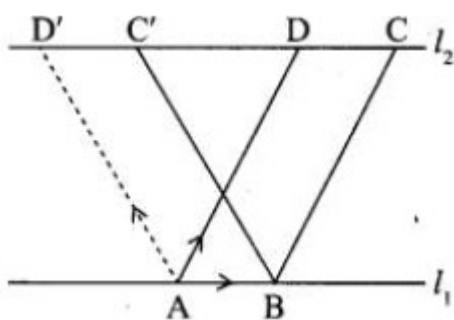
$\therefore$ The area of the quadrilateral ABCD = $12|\vec{AC} \times \vec{BD}|$

Question 7.

Prove by vector method that the parallelogram on the same base and between the same parallels are equal in area.

Solution:

Let ABCD and ABC'D' be two parallelogram between the parallels with same base To prove: Area of ABCD = Area of ABC'D'



$$\begin{aligned} \text{Area of } ABC'D' &= |\vec{AB} \times \vec{AD'}| = |\vec{AB} \times (\vec{AD} + \vec{DD'})| = ||(\vec{AB} \times \vec{AD}) + (\vec{AB} \times \vec{DD'})|| \\ &= |\vec{AB} \times \vec{AD}| + 0 = \text{Area of } ABCD \end{aligned}$$

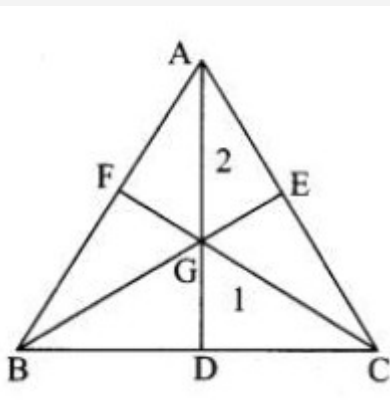
$= |\vec{AB} \times \vec{AD}| + 0$ since $(\vec{AB}$ and $\vec{DD'}$ are parallel).

Question 8.

If G is the centroid of a $\triangle ABC$, prove that.

$$(\text{area of } \triangle GAB) = (\text{area of } \triangle GBC) = (\text{area of } \triangle GAC) = \frac{1}{3} [\text{area of } \triangle ABC]$$

Solution:



$$\begin{aligned} \text{Area of } \triangle GAB &= 12(\vec{AB} \times \vec{AG}) = 12[(\vec{OB} - \vec{OA}) \times (\vec{OG} - \vec{OA})] = 12\{(b^2 - a^2) \times [a \times b + c^3 - a]\} \\ &= 12\{(b^2 - a^2) \times [b \times c - 2a^3]\} = 16\{b \times c - 2b \times a - a \times b - a \times c\} = 16\{b \times c + c \times a + a \times b\} = 13 \times 12\{b \times c + c \times a + a \times b\} = 13 [\text{area of } \triangle ABC] \end{aligned}$$

Similarly we can prove

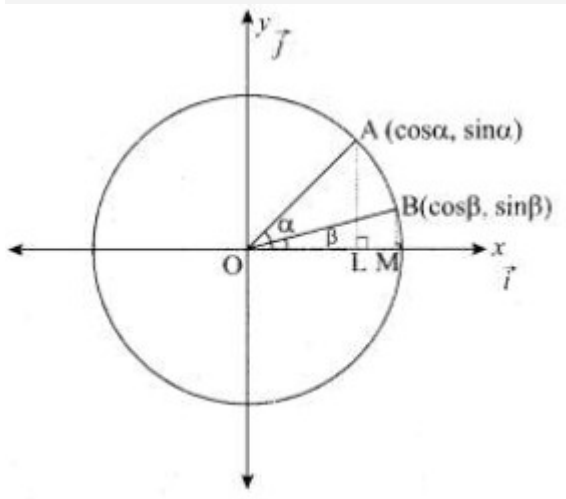
$$\text{Area of } \triangle GBC = \text{Area of } \triangle GAC = \frac{1}{3} [\text{Area of } \triangle ABC]$$

Question 9.

Using vector method, prove that $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$.

Solution:

- Take two points A and B on the unit circle with centre as origin 'O'. so



$$|\vec{OA}| = |\vec{OB}| = 1$$

$$\angle AOX = \alpha; \angle BOX = \beta \Rightarrow \angle AOB = \alpha - \beta$$

- Let $\vec{i}$ and $\vec{j}$ be the unit vectors along the x, y respectively.

- The co-ordinates of A and B be $(\cos\alpha, \sin\alpha)$ and $(\cos\beta, \sin\beta)$ respectively.

$$\vec{OA} = \vec{OL} + \vec{LA} \Rightarrow \vec{OA} = \cos\alpha\vec{i} + \sin\alpha\vec{j} \quad \vec{OB} = \vec{OM} + \vec{MB} \Rightarrow \vec{OB} = \cos\beta\vec{i} + \sin\beta\vec{j}$$

$$\text{So, } \vec{OB} \cdot \vec{OA} = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

$$\text{But } \vec{OB} \cdot \vec{OA} = |\vec{OB}| |\vec{OA}| \cos(\alpha - \beta)$$

$$\vec{OB} \cdot \vec{OA} = \cos(\alpha - \beta)$$

From (1) and (2), we get

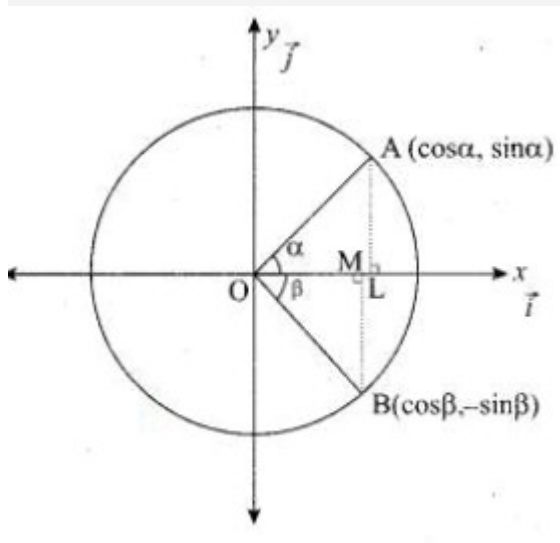
$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

Question 10.

Prove by vector method that $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$.

Solution:

Take two points A and B on the unit circle with centre as origin 'O', so $|\vec{OA}| = |\vec{OB}| = 1$



$$\angle AOX = \alpha; \angle BOX = \beta \Rightarrow \angle AOB = \alpha + \beta$$

Let $\vec{i}$ and $\vec{j}$ be the unit vectors along the x, y direction respectively.

The co-ordinates of A and B be $(\cos\alpha, \sin\alpha)$ and $(\cos\beta, -\sin\beta)$ respectively.

$$\vec{OA} = \vec{OL} + \vec{LA} \Rightarrow \vec{OA} = \cos\alpha\vec{i} + \sin\alpha\vec{j} \quad \vec{OB} = \vec{OM} + \vec{MB} \Rightarrow \vec{OB} = \cos\beta\vec{i} - \sin\beta\vec{j}$$

$$\text{So, } \vec{OB} \times \vec{OA} = (\sin\alpha\cos\beta + \cos\alpha\sin\beta)\vec{k}$$

But, $\vec{OB} \times \vec{OA} = |\vec{OB}| |\vec{OA}| \sin(\alpha + \beta)$

$\vec{OB} \times \vec{OA} = \sin(\alpha + \beta)k$

From (1) & (2), we get

$$\sin(\alpha + \beta) = \sin\alpha \cos\beta + \cos\alpha \sin\beta$$

Question 11.

A particle acted on by constant forces $82\vec{i} - 6\vec{k}$ and $-22\vec{j} - 2\vec{k}$ is displaced from the point (1,2,3) to the point (5,4,1). Find the total work done by the forces.

Solution:

$$\vec{F} = \vec{F}_1 + \vec{F}_2 = (82\vec{i} - 6\vec{k}) + (-22\vec{j} - 2\vec{k}) = (82\vec{i} - 22\vec{j} - 8\vec{k})$$

From (1) & (2), we get

$$\text{Work done by the force} = \vec{F} \cdot \vec{d} = 56 + 8 + 16 = 80 \text{ units.}$$

Question 12 .

Forces of magnitude $5\sqrt{2}$ and $10\sqrt{2}$ units acting in the directions $(3\vec{i} + 4\vec{j} + 5\vec{k})$ and $(6\vec{j} - 8\vec{k})$, respectively, act on a particle which is displaced from the point with position vector $(3\vec{j} - 2\vec{k})$ to the point with position vector $(13\vec{i} + 10\vec{j} - 3\vec{k})$. Find the work done by the forces.

Solution:

$$\vec{a} = 3\vec{i} + 4\vec{j} + 5\vec{k} \text{ (since } |\vec{a}| = |\vec{a}| |\vec{a}|)$$

$$= \sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2} \quad \vec{b} = 6\vec{j} - 8\vec{k} \quad |\vec{b}| = \sqrt{36 + 64} = 10 \quad \vec{a} = 5\sqrt{2} \frac{(3\vec{i} + 4\vec{j} + 5\vec{k})}{\sqrt{50}} = 3\vec{i} + 4\vec{j} + 5\vec{k}$$

$$\vec{b} = 6\vec{j} - 8\vec{k} \quad |\vec{b}| = \sqrt{100 + 64} = \sqrt{164} = 10\sqrt{2} \quad \vec{F}_2 = 10\sqrt{2} \frac{(6\vec{j} - 8\vec{k})}{\sqrt{164}} = 10\sqrt{2} \frac{(3\vec{j} - 4\vec{k})}{\sqrt{41}} = 10\sqrt{2} (3\vec{j} - 4\vec{k})$$

$$\text{So, } \vec{F} = \vec{F}_1 + \vec{F}_2 = (3\vec{i} + 4\vec{j} + 5\vec{k}) + (16\vec{i} + 8\vec{j} - 4\vec{k}) = 19\vec{i} + 12\vec{j} + \vec{k}$$

From (1) & (2), we get

$$\text{Work done by the force} = \vec{F} \cdot \vec{d} = (19\vec{i} + 12\vec{j} + \vec{k}) \cdot (10\vec{i} + 8\vec{j} - 3\vec{k}) = 190 + 96 - 3 = 283 \text{ Units}$$

Question 13.

Find the magnitude and direction cosines of the torque of a force represented by $3\vec{i} + 4\vec{j} - 5\vec{k}$ about the point with position vector $23\vec{i} + 4\vec{k}$ acting through a point whose position vector is $-23\vec{j} - 3\vec{k}$.

Solution:

$$\vec{r} = 23\vec{i} + 4\vec{k} \text{ (through the point)} - (-23\vec{j} - 3\vec{k}) \text{ (about the point)} = 23\vec{i} + 23\vec{j} + 7\vec{k}$$

$$\vec{F} = 3\vec{i} + 4\vec{j} - 5\vec{k} \quad \text{Torque } (\vec{M}) = \vec{r} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 23 & 23 & 7 \\ 3 & 4 & -5 \end{vmatrix}$$

$$= \vec{i}(-115 - 734) - \vec{j}(-115 + 21) + \vec{k}(8 - 15) = -849\vec{i} + 94\vec{j} - 7\vec{k}$$

$$|\vec{M}| = \sqrt{849^2 + 94^2 + 49} = \sqrt{725178} = 851.57 \quad \text{Direction cosines } (3\sqrt{179}, -11\sqrt{179}, -7\sqrt{179})$$

Question 14.

Find the torque of the resultant of the three forces represented by $36\vec{i} - 6\vec{k}$, $4\vec{i} - 10\vec{j} + 12\vec{k}$ and $7\vec{i} + 7\vec{j}$ acting at the point with position vector $8\vec{i} + 4\vec{k}$, about the point with position vector $8\vec{i} - 9\vec{k}$

Solution:

$$\vec{F}_1 = 36\vec{i} - 6\vec{k}, \vec{F}_2 = 4\vec{i} - 10\vec{j} + 12\vec{k}, \vec{F}_3 = 7\vec{i} + 7\vec{j} \quad \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 47\vec{i} - 10\vec{j} + 6\vec{k}$$

$$\vec{r} = (8\vec{i} + 4\vec{k}) - (8\vec{i} - 9\vec{k}) = 13\vec{k} \quad \text{Torque } (\vec{M}) = \vec{r} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 13 \\ 47 & -10 & 6 \end{vmatrix}$$

$$= \vec{i}(-78) - \vec{j}(-85) + \vec{k}(0) = -78\vec{i} + 85\vec{j}$$

$$|\vec{M}| = \sqrt{78^2 + 85^2} = \sqrt{13145} = 114.65$$

Additional Problems

Question 1.

The work done by the force $\vec{F} = a\vec{i} + b\vec{j} + c\vec{k}$ in moving the point of application from (1,1,1) to (2, 2, 2) along a straight

line is given to be 5 units. Find the value of a.

Solution:

$$\vec{F} = a\vec{i} + \vec{j} + \vec{k}; \vec{OA} = \vec{i} + \vec{j} + \vec{k}; \vec{OB} = 2\vec{i} + 2\vec{j} + 2\vec{k}$$

Work done = 5 units

$$\vec{d} = \vec{AB} = \vec{OB} - \vec{OA} = \vec{i} + \vec{j} + \vec{k} \quad \text{Work done} = \vec{F} \cdot \vec{d} = (a\vec{i} + \vec{j} + \vec{k}) \cdot (\vec{i} + \vec{j} + \vec{k}) = 5 = a + 1 + 1 \Rightarrow a = 3$$

Question 2.

If the position vectors of three points A, B and C are respectively $2\vec{i} + 3\vec{j} + 3\vec{k}$, $4\vec{i} + \vec{j} + 5\vec{k}$ and $7(\vec{i} + \vec{k})$. Find $\vec{AB} \times \vec{AC}$. Interpret the result geometrically.

Solution:

$$\vec{OA} = 2\vec{i} + 3\vec{j} + 3\vec{k}; \vec{OB} = 4\vec{i} + \vec{j} + 5\vec{k}; \vec{OC} = 7\vec{i} + 7\vec{k} \quad \vec{AB} = \vec{OB} - \vec{OA} = (4\vec{i} + \vec{j} + 5\vec{k}) - (2\vec{i} + 3\vec{j} + 3\vec{k}) = 2\vec{i} - 2\vec{j} + 2\vec{k}$$

$$\vec{AC} = \vec{OC} - \vec{OA} = (7\vec{i} + 7\vec{k}) - (2\vec{i} + 3\vec{j} + 3\vec{k}) = 5\vec{i} - 3\vec{j} + 4\vec{k}$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 2 \\ 5 & -3 & 4 \end{vmatrix}$$

$$= \vec{i}(2 \cdot 4 - 2 \cdot (-10)) - \vec{j}(2 \cdot 20 - 10) + \vec{k}(-6 - 10)$$

$$= 24\vec{i} - 10\vec{j} - 16\vec{k}$$

The vectors $\vec{AB}$ and $\vec{AC}$ are parallel. But they have the point A as a common point. $\vec{AB}$ and $\vec{AC}$ are along the same line.

$\therefore$ A, B, C are collinear.

Question 3.

A force given by $3\vec{i} - 4\vec{k}$ is applied at the point (1, -1, 2). Find the moment of the force about the point (2, -1, 3).

Solution:

$$\vec{OP} = \vec{i} - \vec{j} + 2\vec{k}; \vec{OA} = 2\vec{i} - \vec{j} + 3\vec{k} \quad \vec{AP} = \vec{OP} - \vec{OA} = (\vec{i} - \vec{j} + 2\vec{k}) - (2\vec{i} - \vec{j} + 3\vec{k}) = -\vec{i} - \vec{k}$$

The moment $\vec{M}$ of the force $\vec{F}$ about the point A is given by

$$\vec{M} = \vec{r} \times \vec{F}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 0 & -1 \\ 3 & 0 & -4 \end{vmatrix}$$

$$= \vec{i}(4 - 0) - \vec{j}(4 - 3) + \vec{k}(0 - 3)$$

$$= 4\vec{i} - \vec{j} - 3\vec{k}$$

Question 4.

Show that the area of a parallelogram having diagonals $3\vec{i} - 2\vec{k}$ and $4\vec{j} + 4\vec{k}$ is $5\sqrt{3}$

Solution:

$$\text{Let } \vec{d}_1 = 3\vec{i} - 2\vec{k} \text{ and } \vec{d}_2 = 4\vec{j} + 4\vec{k}$$

$$\text{Area of the parallelogram} = \frac{1}{2} |\vec{d}_1 \times \vec{d}_2|$$

$$= \frac{1}{2} \left| \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 0 & -2 \\ 0 & 4 & 4 \end{vmatrix} \right|$$

$$= \frac{1}{2} \sqrt{(-12)^2 + (-12)^2 + (12)^2} = \frac{1}{2} \sqrt{300} = 15\sqrt{3}$$

$$\text{Area of the parallelogram} = 15\sqrt{3} \text{ Sq. units.}$$

Ex6.2

Question 1.

If $\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}$, $\vec{b} = 2\vec{i} + \vec{j} - 2\vec{k}$, $\vec{c} = 3\vec{i} + 2\vec{j} + \vec{k}$, find $\vec{a} \cdot (\vec{b} \times \vec{c})$.

Solution:

Given

$$\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}, \vec{b} = 2\vec{i} + \vec{j} - 2\vec{k}, \vec{c} = 3\vec{i} + 2\vec{j} + \vec{k}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{vmatrix}$$

|

$$[\vec{a} \times \vec{c}] = 5\vec{i} - 8\vec{j} + \vec{k} \times 58\vec{k} = 5 \times 16 + 3 = 24 \text{ Aliter: } (\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{c} \cdot (\vec{a} \times \vec{b}) = \vec{c} \cdot \vec{a} \times \vec{b}$$

$$||1-2321-2321||$$

$$||=1(5)+2(8)+3(1)=5+16+3=24$$

Question 2.

Find the volume of the parallelepiped whose coterminous edges are represented by the vectors $-6\vec{i}+14\vec{j}+10\vec{k}$, $14\vec{i}-10\vec{j}-6\vec{k}$ and $2\vec{i}+4\vec{j}-2\vec{k}$.

Solution:

Volume of the parallelepiped $= |[\vec{a}, \vec{b}, \vec{c}]|$

$$[\vec{a} \vec{b} \vec{c}] = ||$$

$$||-6141014-10-624-2||$$

$$||=-6[20+24]-14[-28+12]+10[56+20]=-264+224+760=720 \text{ cubic units}$$

Question 3.

The volume of the parallelepiped whose coterminus edges are $7\vec{i}+2\vec{j}-3\vec{k}$, $7\vec{j}-3\vec{k}$, $-3\vec{i}+7\vec{j}+5\vec{k}$ is 90 cubic units.

Find the value of λ

Solution:

Given, Volume of the parallelepiped = 90 cubic units

$$(\text{i.e.}), |[\vec{a} \vec{b} \vec{c}]| = 90$$

$$||7\lambda-312-1-375||$$

$$||=90 \Rightarrow 7[10+7]-\lambda[5-3]-3[7+6]=90 \Rightarrow 119-2\lambda-39=90 \Rightarrow -2\lambda=10 \Rightarrow \lambda=-10/2=-5$$

Question 4.

If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors represented by concurrent edges of a parallelepiped of volume 4 cubic units, find the value of $(\vec{a}+\vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b}+\vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c}+\vec{a}) \cdot (\vec{a} \times \vec{b})$

Solution:

Let $\vec{a}, \vec{b}, \vec{c}$ be the concurrent edges of parallelepiped

Given volume of parallelepiped = 4 cubic units

$$|[\vec{a}, \vec{b}, \vec{c}]| = 4$$

But,

$$[\vec{a} \vec{b} \vec{c}] = \pm 4$$

$$(\vec{a}+\vec{b}) \cdot (\vec{b} \times \vec{c}) \equiv \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{c}) = [\vec{a} \vec{b} \vec{c}] + [\vec{b} \vec{b} \vec{c}] = [\vec{a} \vec{b} \vec{c}] + 0 = [\vec{a} \vec{b} \vec{c}]$$

$$\text{Similarly } (\vec{b}+\vec{c}) \cdot (\vec{c} \times \vec{a}) \equiv \vec{b} \cdot (\vec{c} \times \vec{a}) + \vec{c} \cdot (\vec{c} \times \vec{a}) = [\vec{b} \vec{c} \vec{a}] + 0 = [\vec{a} \vec{b} \vec{c}]$$

$$\text{So, } (1)+(2)+(3) = 2[\vec{a} \vec{b} \vec{c}] = 2(\pm 4) \Rightarrow \pm 8$$

Question 5.

Find the altitude of a parallelepiped determined by the vectors $\vec{a} = -2\vec{i}+5\vec{j}+3\vec{k}$, $\vec{b} = \vec{i}+3\vec{j}-2\vec{k}$ and $\vec{c} = -3\vec{i}+\vec{j}+4\vec{k}$ if the base is taken as the parallelogram determined by $\vec{b}$ and $\vec{c}$.

Solution:

Volume = Base Area $\times$ Height

$$|[\vec{a}, \vec{b}, \vec{c}]| = |\vec{b} \times \vec{c}| \times \text{Height } |[\vec{a}, \vec{b}, \vec{c}]| = ||$$

$$||-25313-2-314||$$

$$||=-2(12+2)-5(4-6)+3(1+9)=-28+10+30=12 \quad \vec{b} \times \vec{c} = 2\vec{i} + 10\vec{j} + 14\vec{k}$$

|

$$||12\vec{i}+10\vec{j}+14\vec{k}||$$

|

$$||=14|| = \sqrt{14^2 + 10^2 + 12^2} = \sqrt{196 + 100 + 144} = \sqrt{440} = 2\sqrt{110} \quad \text{Height} = 12$$

$$\sqrt{440} = 2\sqrt{110} = 2\sqrt{11 \times 10} = 2\sqrt{11} \times \sqrt{10} \quad \text{Height} = 12$$

Question 6.

Determine whether the three vectors $2\hat{i}+3\hat{j}+\hat{k}$, $\hat{i}-2\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}+3\hat{k}$ are coplanar.

Solution:

$$\text{Let } \vec{a}=2\hat{i}+3\hat{j}+\hat{k}, \vec{b}=\hat{i}-2\hat{j}+2\hat{k}, \vec{c}=3\hat{i}+\hat{j}+3\hat{k}$$

we know that $\vec{a}, \vec{b}, \vec{c}$ are coplanar if and only if $[\vec{a}, \vec{b}, \vec{c}]=0$

$$[\vec{a}, \vec{b}, \vec{c}]=\begin{vmatrix} 2 & 3 & 1 \\ 1 & -2 & 2 \\ 3 & 1 & 3 \end{vmatrix}$$

$$=2(-6-2)-3(3-6)+(1+6)$$

$$=2(-8)-3(-3)+7=-16+9+7=0$$

$\therefore$ Gives vectors are coplanar

Question 7.

Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}$ and $\vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$. If $c_1=1$ and $c_2=2$, find c_3 such that $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar.

Solution:

Given

$$\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=\hat{i}, \vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k} \text{ are coplanar.}$$

But $c_1=1$ and $c_2=2$

So

$$\vec{a}=2\hat{i}+2\hat{j}+\hat{k}$$

We know that $[\vec{a}, \vec{b}, \vec{c}]=0$

$$\begin{vmatrix} 2 & 2 & 1 \\ 2 & 2 & 0 \\ 1 & 1 & c_3 \end{vmatrix}$$

$$=2(2c_3-1)-1(2-2)$$

$$=4c_3-2-0=0 \Rightarrow 4c_3=2 \Rightarrow c_3=\frac{1}{2}$$

Question.

If $\vec{a}=\hat{i}+\hat{k}$, $\vec{b}=x\hat{i}+\hat{j}+(1-x)\hat{k}$, $\vec{c}=y\hat{i}+x\hat{j}+(1+x-y)\hat{k}$, show that $[\vec{a}, \vec{b}, \vec{c}]$ depends neither x nor y .

Solution:

$$\text{Given } \vec{a}=\hat{i}+\hat{k}, \vec{b}=x\hat{i}+\hat{j}+(1-x)\hat{k}, \vec{c}=y\hat{i}+x\hat{j}+(1+x-y)\hat{k}$$

$$[\vec{a}, \vec{b}, \vec{c}]=\begin{vmatrix} 1 & 0 & 1 \\ x & 1 & 1-x \\ y & x & 1+x-y \end{vmatrix}$$

$$=1(1-x-y)-x(1-x)-1(x^2-y)=1-x-y-x+x^2-x^2+y$$

$\therefore$ Clearly $[\vec{a}, \vec{b}, \vec{c}]$ depends on neither x nor y

Question 9.

If the vectors

$$a\hat{i}+a\hat{j}+c\hat{k}, \hat{i}+\hat{k} \text{ and } c\hat{i}+c\hat{j}+b\hat{k}$$

are coplanar, prove that c is the geometric mean of a and b .

Solution:

$$\text{Let } \vec{a}=a\hat{i}+a\hat{j}+c\hat{k}$$

$$\vec{b}=\hat{i}+\hat{k}, \vec{c}=c\hat{i}+c\hat{j}+b\hat{k}$$

But $\vec{a}, \vec{b}, \vec{c}$ are coplanar (Given)

$$\text{So, } [\vec{a}, \vec{b}, \vec{c}]=0$$

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix}$$

$$=a(ac-b)-a(c^2-bc)+c(b-c^2)$$

$$=0 \Rightarrow a(ac-b)-a(c^2-bc)+c(b-c^2)=0 \Rightarrow ac-ab+ac-bc+c^2=0 \Rightarrow c^2=ab \Rightarrow c=\sqrt{ab}$$

$\therefore c$ is the geometric means of ' a ' and ' b '.

Question 10.

Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors such that $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$ show that $[\vec{a}, \vec{b}, \vec{c}]^2=|a|^2|b|^2$.

Solution:

$$11i + 17j - 79k \cdot (a \times (b \times c)) = 91 - k(x)$$

$$11i + 17j - 79k \cdot (a \times (b \times c)) = 91 - k(x)$$

$$11i + 17j - 79k \cdot (a \times (b \times c)) = 91 - k(x)$$

$$(a \cdot b)c = (-2 - 6 - 3)b - (6 + 15 - 2)c = -11b - 19c = -33i - 53j - 22k + 19i + 38j - 57k = -14i - 17j - 79k$$

From (1) & (2), we get

$$a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$$

Question 5.

$$a = 2^i + 3^j - 4^k, b = -i + 2^j - 4^k, c = i + j + k \text{ then find the value of } (a \times b) \cdot (a \times c)$$

Solution:

$$\text{Given } a = 2^i + 3^j - 4^k, b = -i + 2^j - 4^k, c = i + j + k$$

$$(a \times b) \cdot (a \times c) = ||a \cdot a \cdot c \cdot b \cdot a \cdot b \cdot c|| = ||(4 + 9 + 1)(2 + 3 - 1)(-2 + 6 + 4)(-1 + 2 - 4)|| = ||1448 - 3|| = -42 - 32 = -74$$

Question 6.

$$\text{If } \vec{a}, \vec{b}, \vec{c}, \vec{d} \text{ are coplanar vectors, show that } (\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 0$$

Solution:

$$\text{Given } \vec{a}, \vec{b}, \vec{c}, \vec{d} \text{ are coplanar vectors.}$$

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = [\vec{a} \vec{b} \vec{c}] \cdot \vec{d} = 0$$

Question 7.

$$a = i + 2^j + 3^k, b = 2^i - j + k, c = 3^i + 2^j + k \text{ and } a \times (b \times c) = l a + m b + n c, \text{ find the values of } l, m, n$$

Solution:

$$\text{Given } a = i + 2^j + 3^k, b = 2^i - j + k, c = 3^i + 2^j + k$$

$$(a \cdot b)c = (3 + 4 + 3)b - (2 - 2 + 3)c = 10b - 3c = 20i - 10j + 10k - 9i - 6j - 3k = 11i - 16j + 7k$$

From (1) & (2) compare the coefficients.

$$l + 2m + 3n = 11, -m + 2n = -16, l + m + n = 7$$

On solving (3) & (4)

$$(3) \Rightarrow l + 2m + 3n = 11$$

$$(4) \times 2 \Rightarrow 4l - 2m + 4n = -32$$

$$\underline{5l \quad + 7n = -21}$$

$$5l + 7n = -21$$

On solving (4) & (5)

$$(4) \Rightarrow 2l - m + 2n = -16$$

$$(5) \Rightarrow 3l + m + n = 7$$

$$\underline{5l \quad + 3n = -9}$$

$$5l + 3n = -9$$

On solving (6) & (7)

$$(6) \Rightarrow 5l + 7n = -21$$

$$(7) \Rightarrow 5l + 3n = -9$$

$$\underline{(-) \quad (-) \quad (+)}$$

$$4n = -12 \Rightarrow n = -3$$

Substituting $n = -3$ in (6)

$$(6) \Rightarrow 5l + 7(-3) = -21$$

$$5l = 0$$

$$\Rightarrow 5l - 21 = -21$$

$$\Rightarrow l = 0$$

Substitute l & n values in (3)

$$(3) \Rightarrow l + 2m + 3n = 11$$

$$2m = 20$$

$$\Rightarrow 2m - 9 = 11$$

$$m = \frac{20}{2}$$

$$\Rightarrow m = 10$$

$$\therefore l = 0; m = 10; n = -3$$

Question 8.

If $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a} \times (\hat{b} \times \hat{c}) = 12\hat{b}$, find the angle between $\hat{a}$ and $\hat{c}$.

Solution:

$$\text{Given } (\hat{a} \times \hat{b}) \times \hat{c} = 12\hat{b} \Rightarrow (\hat{a} \cdot \hat{c})\hat{b} - (\hat{a} \cdot \hat{b})\hat{c} = 12\hat{b}$$

Since $\hat{b}$ and $\hat{c}$ are non collinear vectors. So, equating corresponding coefficients on both sides.

$$\hat{a} \cdot \hat{c} = 12|\hat{a}||\hat{c}|\cos\theta = 12 \Rightarrow \cos\theta = \frac{12}{12} = 1 \Rightarrow \theta = \cos^{-1}(1) \Rightarrow \theta = 0$$

$\therefore \hat{a}$ makes an angle with $\hat{c}$ is 0

Additional Problems

Question 1.

If $\vec{a} = 3\hat{i} + 2\hat{j} - 4\hat{k}, \vec{b} = 5\hat{i} - 3\hat{j} + 6\hat{k}, \vec{c} = 5\hat{i} - \hat{j} + 2\hat{k}$, find (i) $\vec{a} \times (\vec{b} \times \vec{c})$ (ii) $(\vec{a} \times \vec{b}) \times \vec{c}$ and show that they are not equal.

Solution:

(i)

$$\vec{b} \times \vec{c} =$$

$$|$$

$$| \hat{i} \hat{j} \hat{k} |$$

$$= 20j + 10k, \vec{a} \times (\vec{b} \times \vec{c}) = 00(\times)$$

$$= 32 - 402010$$

$$= 1000i - 30j + 60k$$

$$(ii) \vec{a} \times \vec{b} =$$

$$= 32 - 45 - 36$$

$$= 88j - 19k$$

$$(\vec{a} \times \vec{b}) \times \vec{c} =$$

$$= 38 - 195 - 12$$

$$= 550i - 95j + 190k$$

Question 2 .

. If $\vec{a} = -5\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{j} - 3\hat{k}$ verify that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

Solution:

$$\vec{b} \times \vec{c} =$$

$$= 20501 - 3$$

$$= -5(6) + k(-2) = -3i - 6j - 2k \text{ So, } \vec{a} \times (\vec{b} \times \vec{c}) =$$

$$= 3 - 1 - 5 - 6 - 2$$

$$\begin{aligned} &= (-6 - 6) = j(-4 - 5) + k(-12 + 15) = -12i + 9j + 3k \\ \vec{a} \cdot \vec{c} &= (2i + 3j - k) \cdot (-2i + 5k) = (-2)(2) + (3)(1) + (-1)(-3) = 3 + 3 = 6 \\ \vec{a} \cdot \vec{b} &= (2i + 3j - k) \cdot (-2i + 3k) = (2)(-2) + (3)(0) + (-1)(5) = -4 + 0 - 5 = -9 \\ \text{So, } (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} &= 6[-2i + 3k] - (-9)[j - 3k] = 6[-2i + 3k] + 9[j - 3k] \\ &= -12i + 30k + 9j - 27k = -12i + 9j + 3k \\ (1) &= (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \end{aligned}$$

Ex 6.4

Question 1.

Find the non-parametric form of vector equation and Cartesian equations of the straight line passing through the point with position vector $4\hat{i} + 3\hat{j} - 7\hat{k}$ and parallel to the vector $2\hat{i} - 6\hat{j} + 7\hat{k}$

Solution:

Given

$$\vec{a} = 4\hat{i} + 3\hat{j} - 7\hat{k}, \vec{b} = 2\hat{i} - 6\hat{j} + 7\hat{k}$$

$\therefore$ Non- Parametric vector equation of the straight line is $(\vec{r} - \vec{a}) \times \vec{b} = 0$

$$(4\hat{i} + 3\hat{j} - 7\hat{k}) \times (\vec{r} - 2\hat{i} + 6\hat{j} - 7\hat{k}) = 0$$

Cartesian equation:

$$x - x_1b_1 = y - y_1b_2 = z - z_1b_3 \quad (x_1y_1z_1) = (4-3-7) \quad (b_1b_2b_3) = (2-6-7)$$

Question 2.

Find the parametric form of vector equation and Cartesian equations of the straight line passing through the

point $(-2, 3, 4)$ and parallel to the straight line $x-1-4=y+35=8-z6$

Solution:

Given 234456 , $k4b=-5i+6j+6k$

$\therefore$ Parametric vector equation of the straight line is

$$\vec{r} = \vec{a} + t\vec{b} = (-2i + 3j + 4k) + t(-5i + 6j + 6k), t \in \mathbb{R}$$

Cartesian equation:

$$\frac{x+2}{-5} = \frac{y-3}{6} = \frac{z-4}{6}$$

Question 3.

Find the points where the straight line passes through $(6, 7, 4)$ and $(8, 4, 9)$ cuts the xz and yz planes.

Solution:

Given straight line passing through the points $(6, 7, 4)$ and $(8, 4, 9)$.

Direction ratio of the straight line joining these two points $2, -3, -5$.

Cartesian equation:

$$\frac{x-6}{2} = \frac{y-7}{-3} = \frac{z-4}{-5} = t \text{ (say)}$$

$\therefore$ (i) The straight line cuts the xz -plane.

So we get

$$y = 0 \Rightarrow -3t = -7 \Rightarrow t = \frac{7}{3}$$

The required point $(323, 0, 473)$

(ii) The straight Line cuts yz -plane

So we get $x = 0$

$$2t = 0 \Rightarrow t = -\frac{6}{2} = -3$$

The required point $(0, 16, -11)$.

Question 4.

Find the direction cosines of the straight line passing through the points $(5, 6, 7)$ and $(7, 9, 13)$. Also, find the parametric form of vector equation and Cartesian equations of the straight line passing through two given points.

Solution:

Given straight line passing through the points $(5, 6, 7)$ and $(7, 9, 13)$

$\therefore$ d.r.s: $2, 3, 6$

$$\text{d.c.s} = \left(\frac{2}{\sqrt{2^2+3^2+6^2}}, \frac{3}{\sqrt{2^2+3^2+6^2}}, \frac{6}{\sqrt{2^2+3^2+6^2}} \right) = \left(\frac{2}{7}, \frac{3}{7}, \frac{6}{7} \right)$$

$\therefore$ Parametric form of vector equation passing through two points.

Given

$$\vec{r} = (5i + 6j + 7k) + t(2i + 3j + 6k)$$

Cartesian equation:

$$\frac{x-5}{2} = \frac{y-6}{3} = \frac{z-7}{6}$$

Note: Selection of $\vec{a}$ and $\vec{b}$ is your choice.

Educational Resources

Question 5.

Find the acute angle between the following lines.

$$(i) \vec{r} = (4i - j) + t(i + 2j - 2k), \vec{r} = (i - 2j + 4k) + s(-i - 2j + 2k)$$

$$(ii) x+43=y-74=z+55, \vec{r} = 4k + t(2i + j + k)$$

$$(iii) 2x=3y=-z \text{ and } 6x=-y=-4z$$

Solution:

(i)

$$\text{Given } \vec{b} = i + 2j - 2k, \vec{d} = i - 2j + 2k \Rightarrow \cos \theta = \frac{|\vec{b} \cdot \vec{d}|}{|\vec{b}| |\vec{d}|} = \frac{|(-1 - 4 - 4)|}{\sqrt{1+4+4} \sqrt{1+4+4}} = \frac{9}{3 \times 3} = 1 \Rightarrow \theta = \cos^{-1}(1) = 0^\circ$$

(ii)

$$\text{Given } \vec{b} = 3i + 4j + 5k, \vec{d} = 2i + j + k \Rightarrow \cos \theta = \frac{|\vec{b} \cdot \vec{d}|}{|\vec{b}| |\vec{d}|} = \frac{|6 + 4 + 5|}{\sqrt{9+16+25} \sqrt{4+1+1}} = \frac{15}{5 \times \sqrt{6}} = \frac{3}{\sqrt{6}}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{3}{\sqrt{6}}\right)$$

(iii)

$$\begin{array}{l|l} 2x = 3y = -z & 6x = -y = -4z \\ \frac{x}{1/2} = \frac{y}{1/3} = \frac{z}{-1} & \frac{x}{1/6} = \frac{y}{-1} = \frac{z}{-1/4} \\ \vec{b} = \frac{1}{2}\vec{i} + \frac{1}{3}\vec{j} - \vec{k} & \vec{d} = \frac{1}{6}\vec{i} - \vec{j} - \frac{1}{4}\vec{k} \\ \vec{b} \cdot \vec{d} = \frac{1}{12} - \frac{1}{3} + \frac{1}{4} = \frac{1-4+3}{12} = 0 \end{array}$$

$$\vec{b} \cdot \vec{d} = 112 - 13 + 14 = 1 - 4 + 3 = 0 \Rightarrow \cos \theta = \frac{\vec{b} \cdot \vec{d}}{|\vec{b}| |\vec{d}|} = 0 \Rightarrow \sqrt{(12)^2 + (13)^2 + (-1)^2} \sqrt{(16)^2 + (-1)^2 + (-4)^2} \cos \theta = 0 \Rightarrow \theta = \cos^{-1}(0) \Rightarrow \theta = \pi/2$$

Question 6.

The vertices of $\triangle ABC$ are $A(7,2,1)$, $B(6,0,3)$, and $C(4,2,4)$. Find $\angle ABC$.

Solution:

Let

$$\vec{OA} = 7\vec{i} + 2\vec{j} + \vec{k}, \vec{OB} = 6\vec{i} + 0\vec{j} + 3\vec{k}, \vec{OC} = 4\vec{i} + 2\vec{j} + 4\vec{k} \Rightarrow \vec{AB} = \vec{OB} - \vec{OA} = -\vec{i} - 2\vec{j} + 2\vec{k} \Rightarrow \vec{BC} = \vec{OC} - \vec{OB} = -2\vec{i} + 2\vec{j} + \vec{k}$$

$$\cos \theta = \frac{\vec{AB} \cdot \vec{BC}}{|\vec{AB}| |\vec{BC}|} = \frac{-2 - 4 + 2}{\sqrt{1+4+4} \sqrt{4+4+1}} = \frac{-4}{\sqrt{9} \sqrt{9}} = -\frac{4}{9} \Rightarrow \theta = \cos^{-1}\left(-\frac{4}{9}\right)$$

Question 7.

If the straight line joining the points $(2,1,4)$ and $(a-1,4,-1)$ is parallel to the line joining the points $(0,2,b-1)$ and $(5,3,-2)$, find the values of a and b .

Solution:

$$\vec{OA} = 2\vec{i} + \vec{j} + 4\vec{k}, \vec{OB} = (a-1)\vec{i} + 4\vec{j} - \vec{k}$$

So,

$$\vec{AB} = \vec{OB} - \vec{OA} = (a-3)\vec{i} + 3\vec{j} - 5\vec{k}$$

$$\vec{OC} = 2\vec{i} + (b-1)\vec{k}$$

$$\vec{OD} = 5\vec{i} + 3\vec{j} - 2\vec{k} \Rightarrow \vec{CD} = \vec{OD} - \vec{OC} = 3\vec{i} + \vec{j} - (b-1)\vec{k}$$

Given $\vec{AB}$ and $\vec{CD}$ are parallel.

$$\vec{AB} = m \vec{CD} \Rightarrow (a-3)\vec{i} + 3\vec{j} - 5\vec{k} = m(3\vec{i} + \vec{j} - (b-1)\vec{k})$$

Equating $\vec{i}, \vec{j}, \vec{k}$ coefficient on both side

$$a-3 = 3m, 3 = m, -5 = -(b-1)m \Rightarrow m = 3 \Rightarrow a-3 = 9 \Rightarrow a = 12, -5 = -3(b-1) \Rightarrow -5 = -3b + 3 \Rightarrow -8 = -3b \Rightarrow b = \frac{8}{3}$$

Question 8.

If the straight lines $x-5y+2z=2, y=1, z=1$ and $x=2y+14m=1, z=3$ are perpendicular to each other, find the value of m .

Solution:

$x-5y+2z=2, y=1, z=1$ and $x=2y+14m=1, z=3$ are perpendicular.

Rewrite the above equations.

$$x-5y+2z=2, y=1, z=1 \text{ and } x=2y+14m=1, z=3$$

So, we get,

$$\vec{a} = 2\vec{i} + 3\vec{j} + \vec{k}, \vec{b} = 14m\vec{i} + 3\vec{j} + 3\vec{k} \Rightarrow \vec{a} \cdot \vec{b} = 0 \Rightarrow 2(14m) + 3(3) + 3(1) = 0 \Rightarrow 28m + 12 = 0 \Rightarrow m = -\frac{3}{7}$$

$$\vec{b} = (5m+2)\vec{i} - 5\vec{j} + \vec{k}$$

$$\vec{d} = 2\vec{i} + 3\vec{j} + \vec{k}, \vec{b} \cdot \vec{d} = 0 \Rightarrow 2(2) + 3(3) + 1(1) = 0 \Rightarrow 4 + 9 + 1 = 0 \Rightarrow 14 = 0 \Rightarrow \text{No solution}$$

Question 9.

Show that the points $(2,3,4)$, $(-1,4,5)$ and $(8,1,2)$ are collinear.

Solution:

Given points are $(2, 3, 4)$, $(-1, 4, 5)$ and $(8, 1, 2)$ Equation of the line joining of the first and second point is

$$\frac{x-2}{-3} = \frac{y-3}{1} = \frac{z-4}{1} = m \text{ (say)}$$

$$(-3m+2, m+3, m+4)$$

On putting $m=-2$, we get the third point is $(8,1,2)$

$\therefore$ Given points are collinear.

Additional Problems

Question 1.

Find the vector and cartesian equations of the straight line passing through the point A with position vector $3\mathbf{i}-4\mathbf{j}+4\mathbf{k}$ and parallel to the vector $-5\mathbf{i}+7\mathbf{j}+3\mathbf{k}$.

Solution:

We know that vector equation of the line through the point with position vector $\vec{a}$ and parallel to $\vec{v}$ is given by $\vec{r}=\vec{a}+t\vec{v}$ where t is a scalar.

Here $\vec{a}=3\mathbf{i}-4\mathbf{j}+4\mathbf{k}$ and $\vec{v}=-5\mathbf{i}+7\mathbf{j}+3\mathbf{k}$

Vector equation of the line is

$$\vec{r}=(3\mathbf{i}-4\mathbf{j}+4\mathbf{k})+t(-5\mathbf{i}+7\mathbf{j}+3\mathbf{k})$$

The cartesian equation of the line passing through (x_1, y_1, z_1) and parallel to a vector whose d.r.s are l, m, n is $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$

Here $(x_1, y_1, z_1)=(3, -4, 4)$ and $(l, m, n)=(-5, 7, 3)$

$\therefore$ The required equation is $\frac{x-3}{-5}=\frac{y+4}{7}=\frac{z-4}{3}$

Question 2.

Find the vector and cartesian equations of the straight line passing through the points $(-5, 2, 3)$ and $(4, -3, 6)$.

Solution:

Vector equation of the straight line passing through two points with position vectors $\vec{a}$ and $\vec{b}$ is given by $\vec{r}=\vec{a}+t(\vec{b}-\vec{a})$

Here

$$\vec{a}=-5\mathbf{i}+2\mathbf{j}+3\mathbf{k} \text{ and } \vec{b}=4\mathbf{i}-3\mathbf{j}+6\mathbf{k}$$

$\therefore$ Vector equation of the line is

$$(\text{or}) \vec{r}=(-5\mathbf{i}+2\mathbf{j}+3\mathbf{k})+t(9\mathbf{i}-5\mathbf{j}+3\mathbf{k})$$

Cartesian Form:

The required equation is $\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$

Here $(x_1, y_1, z_1)=(-5, 2, 3); (x_2, y_2, z_2)=(4, -3, 6)$

$\therefore \frac{x+5}{9}=\frac{y-2}{-5}=\frac{z-3}{3}$ is the cartesian equation of the line.

Question 3.

Find the angle between the lines.

$$2\mathbf{i}+2\mathbf{j}+2\mathbf{k} \text{ and } 5\mathbf{i}+2\mathbf{j}+6\mathbf{k}$$

Solution:

Let the given lines be in the direction of $\vec{u}$ and $\vec{v}$

$$\text{Then } \vec{u}=\mathbf{i}+2\mathbf{j}+2\mathbf{k}, \vec{v}=\mathbf{i}+2\mathbf{j}+6\mathbf{k}$$

Let θ be the angle between the given lines

$$(\cos \theta = \frac{(\vec{u} \cdot \vec{v})}{|\vec{u}| |\vec{v}|}) \Rightarrow \cos \theta = \frac{19}{21} \Rightarrow \theta = \cos^{-1} \left(\frac{19}{21} \right)$$

Question 4.

Find the angle between the following lines $x-12=y+13=z-46$ and $x+1=y+22=z-42$

Solution:

Angle between two lines is the same as angle between their parallel vectors.

The parallel vectors are: $23\mathbf{i}+3\mathbf{j}+6\mathbf{k}$ and $12\mathbf{i}+2\mathbf{j}+2\mathbf{k}$.

If θ is the angle between $\vec{a}$ and $\vec{b}$, then $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$.

$$\vec{a} \cdot \vec{b} = (2)(1) + (3)(2) + (6)(2) = 2 + 6 + 12 = 20$$

$$|\vec{a}| = \sqrt{4+9+36} = 7, |\vec{b}| = \sqrt{1+4+4} = 3$$

So, $\cos \theta = \frac{20}{7 \times 3} = \frac{20}{21}$

Ex 6.5

Question 1.

Find the parametric form of vector equation and Cartesian equations of a straight line passing through $(5, 2, 8)$ and is perpendicular to the straight lines $\vec{r}=(\mathbf{i}+\mathbf{j}+\mathbf{k})+s(2\mathbf{i}-2\mathbf{j}+\mathbf{k})$ and $\vec{r}=(2\mathbf{i}-\mathbf{j}-3\mathbf{k})+t(\mathbf{i}+2\mathbf{j}+2\mathbf{k})$

Given point

52 8+5j+2k9z212xj-+k2r, i+=j+2b 2kd x p|82222

$$||^i{}^j{}^k{}^2-21122||$$

$$||636-3j+6k$$

$\therefore$ This' vector is perpendicular to both the given straight lines.

$\therefore$ The required straight line is $\vec{r} = \vec{a} + t(\vec{b} \times \vec{d})$

Cartesian equation:

$$\vec{r} = (3i + 2j + 8k) + t(2i + j - 2k), t \in \mathbb{R} \quad x - 5 = y - 2 = z - 8 \quad (\text{OR}) \quad x - 5 = y - 2 = z - 8$$

Question 2.

Show that the lines $\vec{r} = (6\hat{i} + \hat{j} + 2\hat{k}) + s(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (3\hat{i} + 2\hat{j} - 2\hat{k}) + t(2\hat{i} + 4\hat{j} - 5\hat{k})$ are skew lines and hence find the shortest distance between them.

Given

$$65-16j+2fk-2i-2c-3k-3j-2k-2i=i+2j-4c-5k=-3j-fk(4c-a-d)e \quad |22332224534$$

||-31-412-324-5||

$$|| = -3(-10+12) - 1(-5+6) - 4(4-4) = -6 - 1 \Rightarrow -7 \neq 0$$

Given lines are skew lines

$$\vec{b} \times \vec{d} = ||$$

||ijk|2-324-5||

$$|| = (-10+12)\hat{i} + (-5+6)\hat{j} + (4-4)\hat{k} = 2\hat{i} + \hat{j} \quad ||\vec{x}\vec{d}|| = \sqrt{4+1} = \sqrt{5}$$

Shortest distance between skew lines = $|\vec{r}(c-a)\vec{b}d|\sqrt{5} = 7\sqrt{5}$ units.

Question 3.

If the two lines $x-12=y+13=z-14$ and $x-31=y-mz=Z$ intersect at a point, find the value of m .

Solution:

$$x-12=y+13=z-14=s \text{ (say) } (2s+1, 3s-1, 4s+1) \quad x-31=y-m2=z+t \text{ (say) } (t+3, 2t+m, t)$$

Given two lines are intersecting lines. So, equate the corresponding co-ordinates.

$$2s+1=t+3 \quad 2s-t=2 \dots (1) \quad | 4s+1=t \dots (2) \quad | 3s-1=2t+m$$

Substitute (2) in (1)

$$2s - 4s - 1 = 2 - 2s = 2 + 1 \Rightarrow s = -3/2 \text{ sub } s = -3/2 \text{ in (1)}$$

$$(1) \Rightarrow 2(-32) - t = 2$$

$$-t = 2 + 3 \Rightarrow t = -5$$

Substitute $s=-32$ and $t=-5$ in (3)

$$(3) \Rightarrow 3(-32) - 1 = 2(-5) + m - 92 - 1 = -10 + m \Rightarrow -92 + 9 = m \quad 92 = m$$

Question 4.

Show that the lines $x-33=y-3, z-1=0$ and $x-62=z-13, y-2=0$ intersect. Also find the point of intersection

Solution:

$$x-33=y-3-1, z-1=0 \Rightarrow z=1, x-62=z-13, y-2=0 \Rightarrow y=2, (x_1, y_1, z_1)=(3, 3, 1) \text{ and } (x_2, y_2, z_2)=(6, 2, 1) (b_1, b_2, b_3)=(3, -1, 0) \text{ and } (d_1, d_2, d_3)=(2, 0, 3)$$

Condition for intersection of two lines

$$\begin{pmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{pmatrix}$$

$||=0||$

||3-103-10203||

$\parallel=0$ Since $(R_1 \equiv R_2)$

$\therefore$ Given two lines are intersecting lines.

Any point on the first line

$$x-3z=y-3=1 \text{ and } z=1$$

$$(3\lambda+3, -\lambda+3, 1)$$

Any point on the Second line

$$x-6z=z-13=\mu \text{ and } y=2(2\mu+6, 2, 3\mu+1)$$

$\therefore$ The required point of intersection is $(6, 2, 1)$

Question 5.

Show that the straight lines $x+1=2y=-12z$ and $x=y+2=6z-6$ are skew and hence find the shortest distance between them.

Solution:

$$x+1=2y=-12z \Rightarrow x=y+2=6z-6 \Rightarrow x+1=y+2=z-1/12 \Rightarrow (x_1, y_1, z_1)=(-1, 0, 0) \text{ and } (x_2, y_2, z_2)=(0, -2, 1)$$

$$(b_1, b_2, b_3)=(1, 1/2, -1/12) \text{ and } (d_1, d_2, d_3)=(1, 1, 1/6)$$

Condition for skew lines

$$\parallel$$

$$\parallel x_2-x_1, y_2-y_1, z_2-z_1, b_1b_2b_3d_1d_2d_3 \parallel$$

$$\neq 0 \text{ (OR)} [(c-a) \cdot \vec{bd}] \neq 0 \text{ OR } [(c-a) \cdot \vec{bd}] = -b \cdot (-b) \parallel$$

$$\parallel$$

$$\parallel 1-2, 11-12, 11-11, 16 \parallel$$

$$\parallel$$

$$\parallel = 1[112+112] + 2[16+112] + 1[1-12] = 16+26+212+1-12 = 16+26+16+1-12 = 23+12 = 76 \neq 0$$

Given Lines are skew lines.

$$\vec{b} \times \vec{d} = \parallel$$

$$\parallel$$

$$\parallel$$

$$\parallel i, j, k \parallel 12-11, 21-11, 16 \parallel$$

$$\parallel$$

$$\parallel$$

$$\parallel = (16) = j(312) + k(12) = 16i - 14j + 12k \parallel \vec{b} \times \vec{d} \parallel = \sqrt{16^2 + 14^2 + 12^2} = \sqrt{16+36+144} = \sqrt{196} = 14$$

Shortest distance between skew lines $= \frac{|[(c-a) \cdot \vec{bd}]|}{\parallel \vec{b} \times \vec{d} \parallel} = \frac{76}{14} = \frac{38}{7}$ units.

Question 6.

Find the parametric form of vector equation of the straight line passing through $(-1, 2, 1)$ and parallel to the straight line $\vec{r} = (2\hat{i} + 3\hat{j} - \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k})$ and hence find the shortest distance between the lines.

Solution:

$$\text{Given point } (-1, 2, 1) = -\hat{i} + 2\hat{j} + \hat{k}$$

$$\vec{r} = (-\hat{i} + 2\hat{j} + \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k})$$

Parallel to the straight line

$$\vec{r} = (-\hat{i} + 2\hat{j} + \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k}) \text{ Here } \vec{a} = -\hat{i} + 2\hat{j} + \hat{k}, \vec{b} = \hat{i} - 2\hat{j} + \hat{k}$$

$$\parallel$$

$$\parallel i, j, k \parallel 31-21, 21-21, 21 \parallel$$

$$\parallel$$

$$\parallel = 5i - 5j - 7k \parallel (c-a) \times b \parallel = \sqrt{5^2 + 5^2 + 7^2} = \sqrt{83} \parallel b \parallel = \sqrt{1+4+1} = \sqrt{6}$$

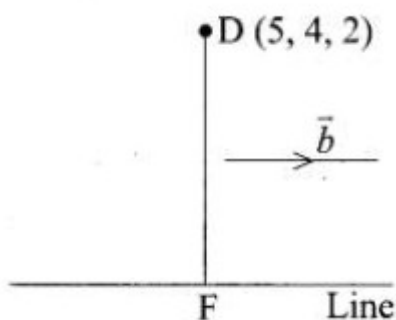
Shortest distance between parallel lines $= \frac{|[(c-a) \times b] \cdot b|}{\parallel (c-a) \times b \parallel \parallel b \parallel} = \frac{57}{\sqrt{83} \sqrt{6}}$ units.

Question 7.

Find the foot of the perpendicular drawn from the point $(5, 4, 2)$ to the line $x+12=y-33=z-1$. Also, find the equation of the perpendicular.

Solution:

Compare the given equation $x+12=y-33=z-1-1$ in to $x-x_1b_1=y-y_1b_2=z-z_1b_3$ Here $\vec{b}=b_1\vec{i}+b_2\vec{j}+b_3\vec{k}$ So, $\vec{b}=2\vec{i}+3\vec{j}-\vec{k}$
 Any point on the line
 $x+12=y-33=z-1-1=t$ (say)
 The point $F(2t-1, 3t+3, -t+1)$



$$\vec{DF} = \vec{OF} - \vec{OD} = (2t-6)\vec{i} + (3t-1)\vec{j} + (-t-1)\vec{k}$$

Since $\vec{b}$ is perpendicular to $\vec{DF}$, we have

$$\vec{b} \cdot \vec{DF} = 0 \Rightarrow 2(2t-6) + 3(3t-1) - 1(-t-1) = 0 \Rightarrow 4t-12+9t-3+t+1=0 \Rightarrow 14t-14=0 \Rightarrow t=1$$

The co-ordinate of F is (1,6,0). The equation of the perpendicular is

$$x-x_1x_2-x_1=y-y_1y_2-y_1=z-z_1z_2-z_1 \text{ and } D(5,4,2), F(1,6,0) \Rightarrow x-5-4=y-42=z-2-2$$

Additional Problems

Question 1.

Find the shortest distance between the parallel line

$$\vec{r} = (i-j) + t(2i-j+k) \text{ and } \vec{r} = (2i-j+k) + s(2i-j+k)$$

Solution:

Compare the given equations with $\vec{r} = a_1 + t\vec{u}$ and $\vec{r} = a_2 + s\vec{u}$,

$$\vec{a}_1 = i-j, \vec{a}_2 = 2i-j+k \text{ and } \vec{u} = 2i-j+k \Rightarrow \vec{a}_2 - \vec{a}_1 = i+2j+k \Rightarrow (\vec{a}_2 - \vec{a}_1) \times \vec{u} =$$

$$\begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 2 & -1 & 1 \end{vmatrix} = i(2-2) - j(1-2) + k(1-4) = j - 3k$$

$$|\vec{a}_2 - \vec{a}_1| = \sqrt{1+4+1} = \sqrt{6} \quad |\vec{u}| = \sqrt{4+1+1} = \sqrt{6}$$

$$|\vec{u} \times (\vec{a}_2 - \vec{a}_1)| = \sqrt{9+1+25} = \sqrt{35} \Rightarrow \text{The Distance between the parallel lines} = \frac{|\vec{u} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{u}|} = \frac{\sqrt{35}}{\sqrt{6}}$$

Question 2.

Show that the two lines $\vec{r} = (i-j) + t(2i-j+k)$ and $\vec{r} = (2i-j) + s(i-j-k)$ skew lines and find the distance between them.

Solution:

Compare the given equations with $\vec{r} = a_1 + t\vec{u}$ and $\vec{r} = a_2 + s\vec{v}$

$$\vec{a}_1 = i-j, \vec{a}_2 = 2i-j \text{ and } \vec{u} = 2i-j+k, \vec{v} = i-j-k$$

$$\vec{a}_2 - \vec{a}_1 = i \Rightarrow (\vec{a}_2 - \vec{a}_1) \times \vec{u} \times \vec{v} =$$

$$\begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 2 & -1 & 1 \end{vmatrix} = i(0-0) - j(1-2) + k(-1-2) = j - 3k$$

$$|\vec{a}_2 - \vec{a}_1| = 1 \neq 0$$

$\therefore$ They are skew lines.

$$\vec{u} \times \vec{v} =$$

$$\begin{vmatrix} i & j & k \\ 2 & -1 & 1 \\ 1 & -1 & -1 \end{vmatrix} = i(1-1) - j(-2-1) + k(2-1) = 3j + k$$

$$|\vec{a}_2 - \vec{a}_1| \times |\vec{u} \times \vec{v}| = 1 \times \sqrt{10} = \sqrt{10}$$

Shortest distance between the lines $= \frac{|\vec{a}_2 - \vec{a}_1| \times |\vec{u} \times \vec{v}|}{|\vec{u} \times \vec{v}|}$

From (1) shortest distance between them is $\sqrt{14}$

Question 3.

Show that the lines $x-13=y-1=z+10$ and $x-42=y-0=z+13$ intersect and hence find the point of intersection

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Solution:

The condition for intersecting is $||$

$$||x_2-x_1 \ y_2-y_1 \ z_2-z_1|| \begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}$$

$$||=0$$

Compare with $x-x_1l_1=y-y_1m_1=z-z_1n_1$ and $x-x_2l_2=y-y_2m_2=z-z_2n_2$, we get

$$(x_1, y_1, z_1) = (1, 1, -1); (x_2, y_2, z_2) = (4, 0, -1); (l_1, m_1, n_1) = (3, -1, 0); (l_2, m_2, n_2) = (2, 0, 3)$$

The determinant becomes

$$||$$

$$||3-103-10203||$$

$$||=0$$

Note that $\vec{r}_u$ and $\vec{r}_v$ are not parallel.

$\therefore$ The lines are intersecting lines.

Point of intersection:

Take

$$x-13=y-1=z+10=\lambda$$

$\therefore$ Any point on the line is of the form $(3\lambda+1, -\lambda+1, -1)$

Take

$$x-42=y-0=z+13=\mu$$

Any point on this line is of the form $(2\mu+4, 0, 3\mu-1)$ Since they are intersecting, for some λ, μ

$$(3\lambda+1, -\lambda+1, -1) = (2\mu+4, 0, 3\mu-1) \Rightarrow \lambda=1 \text{ and } \mu=0$$

To find the point of intersection either take $\lambda=1$ or $\mu=0$

$\therefore$ The point of intersection is $(4, 0, -1)$,

Question 4.

Find the shortest distance between the skew lines.

$$\vec{r} = (i + j) + \lambda(2i + j + k) \text{ and } \vec{r} = (i + j + k) + \mu(2i + j + k)$$

Solution:

Compare the given equation with $\vec{r} = a_1 + t\vec{u}$ and $\vec{r} = a_2 + s\vec{v}$

$$\vec{a}_1 = i + j; \vec{a}_2 = i + j + k; \vec{u} = 2i + j + k; \vec{v} = 2i + j + k \text{ and } \vec{u} \times \vec{v} = 4j - 4k \quad [(\vec{a}_2 - \vec{a}_1) \cdot \vec{u} \times \vec{v}] = ||$$

$$||02-12112-11||$$

$$||=12|\vec{u} \times \vec{v}| = 4\sqrt{2} \quad \text{distance} = \frac{|[(\vec{a}_2 - \vec{a}_1) \cdot \vec{u} \times \vec{v}]|}{|\vec{u} \times \vec{v}|} = \frac{124\sqrt{2}}{4\sqrt{2}} = 3\sqrt{2}$$

Question 5.

Find the shortest distance between the parallel lines.

$$(2i + 3j) + t(2i + 3j) \text{ and } (2i + 3j) + s(2i + 3j)$$

Solution:

Comparing the given equations with $\vec{r} = a_1 + t\vec{u}$ and $\vec{r} = a_2 + s\vec{u}$ we get,

$$\vec{a}_1 = 2i + 3j; \vec{a}_2 = 2i + 3j; \vec{u} = 2i + 3j; \vec{u} = 2i + 3j \text{ and } \vec{u} \times \vec{u} = (2i + 3j) \times (2i + 3j) = (-0j + 0k) \cdot (0i + 0j) = ||$$

$$||$$

$$||i + j - 23 - 1 - 12||$$

$$||$$

$$||=4\sqrt{3}||[2+3] + k[-1-2] = -i - 5j - 3k \quad |\vec{u} \times (\vec{a}_2 - \vec{a}_1)| = \sqrt{1+25+9} = \sqrt{35} \quad |\vec{u}| = \sqrt{1+4+9} = \sqrt{14}$$

The distance between the parallel lines $= \frac{|\vec{u} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{u}|} = \frac{\sqrt{35}\sqrt{14}}{\sqrt{14}} = \sqrt{35}$ units.

Ex 6.6

Question 1.

Find a parametric form of vector equation of a plane which is at a distance of 7 units from the origin having 3, -4, 5 as direction ratios of a normal to it.

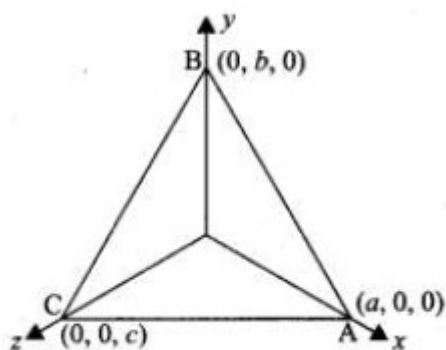
Solution:

Equate these two centroids we get

$$a^3 = 3ub^3 = 3vc^3 = 3wa^3 = 3ub^3 = 3vc^3 = 3wa^3$$

We know that intercept form the plane is

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$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

Additional Problems

Question 1.

Find the vector and cartesian equations of a plane which is at a distance of 18 units from the origin and which is normal to the vector $2i + 7j + 8k$

Solution:

$$|\vec{r} \cdot \vec{n}| = |\vec{r} \cdot (2i + 7j + 8k)| = 18 \Rightarrow \vec{r} \cdot (2i + 7j + 8k) = \pm 18\sqrt{117}$$

Hence, the required vector equation of the plane is $\vec{r} \cdot \vec{n} = p$.
i.e.

$$\vec{r} \cdot (2i + 7j + 8k) = \pm 18\sqrt{117} \text{ i.e. } \vec{r} \cdot (2i + 7j + 8k) = 54\sqrt{13}$$

Taking $\vec{r} = xi + yj + zk$ we get the cartesian equation as

$$(xi + yj + zk) \cdot (2i + 7j + 8k) = 54\sqrt{13}$$

i.e.

$$2x + 7y + 8z = 54\sqrt{13}$$

Question 2.

Find the unit vector to the plane $2x - y + 2z = 5$.

Solution:

Writing the plane in normal form we get,

$$(2i - j + 2k) \cdot (2i - j + 2k) = 5 \text{ i.e. } \vec{r} \cdot (2i - j + 2k) = \frac{5}{\sqrt{4+1+4}} = \frac{5}{3}$$

So, the unit vector normal to the plane is $\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{2i - j + 2k}{3}$

Question 3.

Find the length of the perpendicular from the origin to the plane $\vec{r} \cdot (3i + 4j + 12k) = 26$.

Solution:

Taking the equation of the plane in cartesian form we get,

$$(3x + 4y + 12z) = 26$$

$$\text{i.e., } 3x + 4y + 12z - 26 = 0$$

The length of the perpendicular from (0,0,0) to the above plane is $\pm \frac{-26}{\sqrt{9+16+144}} = \pm \frac{26}{13} = 2$ units

Question 4.

The foot of the perpendicular drawn from the origin to a plane is (8, -4, 3). Find the equation of the plane.

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Solution:

The required plane passes through the point A(8, -4, 3) and is perpendicular to $\vec{OA}$.

$$\vec{OA} = 8i - 4j + 3k \text{ and } \vec{n} = 8i - 4j + 3k$$

The required equation of the plane is $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

$$(8i - 4j + 3k) \cdot (8i - 4j + 3k) = 64 + 16 + 9 = 89$$

The cartesian equation is $8x - 4y + 3z = 89$.

Question 5.

Find the equation of the plane through the point whose position vector is $2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and perpendicular to the vector $4\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$.

Solution:

The required plane is perpendicular to $4\mathbf{i} - 2\mathbf{j} - 3\mathbf{k}$

So, it is parallel to the plane $4x + 2y - 3z = k$

$\therefore$ the equation of the plane is $4x + 2y - 3z = k$

The plane passes through the point $(2, -1, 1)$

$\Rightarrow (4)(2) + 2(-1) - 3(1) = \lambda$ i.e. $\lambda = 8 - 2 - 3 = 3$

So, the equation of the plane is $4x + 2y - 3z = 3$.

Question 6.

Find the vector and cartesian equations of the plane passing through the point $(2, -1, 4)$ and parallel to the plane $\vec{r} \cdot (4\mathbf{i} - 12\mathbf{j} - 3\mathbf{k}) = 7$

Solution:

The given plane is $\vec{r} \cdot (4\mathbf{i} - 12\mathbf{j} - 3\mathbf{k}) = 7$

i.e. $(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot (4\mathbf{i} - 12\mathbf{j} - 3\mathbf{k}) = 7$

i.e. $4x - 12y - 3z = 7$

The required plane is parallel to the above plane. So, the equation of the required plane is $4x - 12y - 3z = k$. The plane passes through $(2, -1, 4)$.

$\Rightarrow 4(2) - 12(-1) - 3(4) = k$ i.e. $k = 8 + 12 - 12 = 8$

So, the equation of the plane is $4x - 12y - 3z = 8$.

Ex 6.7

Question 1.

Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point $(2, 3, 6)$ and parallel to the straight lines $x - 12 = y + 13 = z - 31$ and $x + 32 = y - 3 = z + 1 - 3$

Solution:

The required plane passing through the point $\vec{a} = 2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$ and parallel to the vectors $\vec{b} = 2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\vec{c} = 2\mathbf{i} - 5\mathbf{j} - 3\mathbf{k}$

$\vec{b} \times \vec{c} =$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 1 \\ 2 & -5 & -3 \end{vmatrix}$$

$$= \mathbf{i}(-9 - 3) - \mathbf{j}(-6 - 2) + \mathbf{k}(-10 - 6) = -4\mathbf{i} + 8\mathbf{j} - 16\mathbf{k}$$

Non-parametric form of vector equation

$$(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{c}) = 0 \quad (\text{or}) \quad \vec{r} \cdot (\vec{b} \times \vec{c}) = \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$\vec{r} \cdot (-4\mathbf{i} + 8\mathbf{j} - 16\mathbf{k}) = (2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}) \cdot (-4\mathbf{i} + 8\mathbf{j} - 16\mathbf{k}) = -8 + 24 - 96 = -80$$

$$\vec{r} \cdot (-4\mathbf{i} + 8\mathbf{j} - 16\mathbf{k}) = -80$$

Cartesian equation

$$(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \cdot (-4\mathbf{i} + 8\mathbf{j} - 16\mathbf{k}) = -80$$

$$-4x + 8y - 16z = -80$$

$$x - 2y + 4z = 20$$

Question 2.

Find the parametric form of vector equation, and Cartesian equations of the plane passing through the points $(2, 2, 1)$, $(9, 3, 6)$ and perpendicular to the plane $2x + 6y + 6z = 9$.

Solution:

The required plane passes through the points

$$\vec{a} = 2\mathbf{i} + 2\mathbf{j} + \mathbf{k}, \vec{b} = 9\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}$$

and parallel to the vector $\vec{c} = 2\mathbf{i} + 6\mathbf{j} + 6\mathbf{k}$

$$\vec{b} - \vec{a} = (9\mathbf{i} + 3\mathbf{j} + 6\mathbf{k}) - (2\mathbf{i} + 2\mathbf{j} + \mathbf{k}) = 7\mathbf{i} + \mathbf{j} + 5\mathbf{k}$$

Non-parametric form of vector equation

$$(\vec{r} - \vec{a}) \cdot ((\vec{b} - \vec{a}) \times \vec{c}) = 0 \quad (\vec{b} - \vec{a}) \times \vec{c} =$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 7 & 1 & 5 \\ 2 & 6 & 6 \end{vmatrix}$$

$$\vec{H} = (-4-30)\vec{i} + j(42-10) + k(42-2) = -24\vec{i} - 32\vec{j} + 40\vec{k}$$

$$(1) \Rightarrow (0-420) + 2201232^j + 40^k = 0 \Rightarrow (-24\vec{i} - 32\vec{j} + 40\vec{k}) \cdot \vec{r} = (-24\vec{i} - 32\vec{j} + 40\vec{k}) \cdot (2\vec{i} + 2\vec{j} + \vec{k}) = (-24\vec{i} - 32\vec{j} + 40\vec{k}) \cdot (-24\vec{i} - 32\vec{j} + 40\vec{k}) = -48 - 64 + 40 - 8[r \cdot (3\vec{i} + 4\vec{j} - 5\vec{k})] = -72r \cdot (3\vec{i} + 4\vec{j} - 5\vec{k}) = 9$$

Cartesian equation

$$(3\vec{i} + 4\vec{j} - 5\vec{k}) \cdot (3\vec{i} + 4\vec{j} - 5\vec{k}) = 9 \Rightarrow 3x + 4y - 5z = 9$$

Question 3.

Find the parametric form of vector equation and Cartesian equations of the plane passing through the points (2,2,1), (1,-2,3) and parallel to the straight line passing through the points (2,1,-3) and (-1,5,-8).

Solution:

Equation of the straight line passing through the points (2,1,-3) and (-1,5,-8) is

$$x-2 = y-1 = z+3$$

☐ The required plane passing through the points

$$\vec{a} = 2\vec{i} + 2\vec{j} + \vec{k} \text{ and } \vec{b} = \vec{i} - 2\vec{j} + 3\vec{k}$$

and parallel to $\vec{d} = 3\vec{i} + 4\vec{j} - 5\vec{k}$

Parametric form of vector equation

$$\vec{r} = \vec{a} + s(\vec{b} - \vec{a}) + t\vec{d} = (2\vec{i} + 2\vec{j} + \vec{k}) + s(\vec{i} - 4\vec{j} + 2\vec{k}) + t(-3\vec{i} + 4\vec{j} - 5\vec{k}), s, t \in \mathbb{R}$$

Cartesian equation:

||

$$||x - x_1 y - y_1 z - z_1 x_2 - x_2 y_2 - y_2 z_2 - z_2 x_3 - x_3 y_3 - y_3 z_3 - z_3 x_1||$$

$$||=0||$$

$$||x - 2y - 2z - 1 - 42 - 34 - 5||$$

$$||=0(x-2)[20-8] - (y-2)[5+6] + (z-1)$$

$$[-4-12] = 012(x-2) - 11(y-2) - 16(z-1) = 012x - 24 - 11y + 22 - 16z + 16 = 012x - 11y - 16z + 14 = 0$$

Question 4.

Find the non-parametric form of vector equation of the plane passing through the point (1,-2,4) and perpendicular to the plane $x+2y-3z=11$ and parallel to the line $x+7z=y+3-1=z1$

Solution:

The required plane passing through the point $\vec{a} = \vec{i} - 2\vec{j} + 4\vec{k}$ and parallel to the plane $\vec{b} = \vec{i} + 2\vec{j} - 3\vec{k}$

and parallel to the line $\vec{d} = 3\vec{i} - \vec{j} + \vec{k}$

$$\vec{b} \times \vec{c} = ||$$

|

$$||\vec{i} + 2\vec{j} - 3\vec{k} - 11||$$

|

$$\vec{H} = (-3-30)\vec{i} + j(1+9) + k(-1-6) \Rightarrow \vec{b} \times \vec{c} = -\vec{i} - 10\vec{j} - 7\vec{k}$$

Non-parametric form of vector equation

$$(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{c}) = 0 \Rightarrow (\vec{r} - \vec{a}) \cdot (-\vec{i} - 10\vec{j} - 7\vec{k}) = 0 \Rightarrow \vec{r} \cdot (-\vec{i} - 10\vec{j} - 7\vec{k}) = \vec{a} \cdot (-\vec{i} - 10\vec{j} - 7\vec{k}) = (-1 - 20 - 28) = -29$$

$$(-1 - 20 - 28) = -29 \Rightarrow \vec{r} \cdot (-\vec{i} - 10\vec{j} - 7\vec{k}) = -29 \text{ (or) } \vec{r} \cdot (\vec{i} + 10\vec{j} + 7\vec{k}) = 29$$

Cartesian equation

$$(x+1) + 10(y+2) + 7(z+3) = 29 \Rightarrow x + 10y + 7z - 9 = 0$$

Question 5.

Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r} = (\vec{i} - \vec{j} + 3\vec{k}) + t(2\vec{i} - \vec{j} + 4\vec{k})$ and perpendicular to the plane $\vec{r} \cdot (\vec{i} + 2\vec{j} + \vec{k}) = 8$.

Solution:

The required plane passing through the point $\vec{a} = \vec{i} - \vec{j} + 3\vec{k}$ and parallel to $\vec{b} = 2\vec{i} - \vec{j} + 4\vec{k}$ and $\vec{c} = \vec{i} + 2\vec{j} + \vec{k}$

Parametric form of vector equation

$$\vec{r} = \vec{a} + s\vec{b} + t\vec{c} = (\vec{i} - \vec{j} + 3\vec{k}) + s(2\vec{i} - \vec{j} + 4\vec{k}) + t(\vec{i} + 2\vec{j} + \vec{k}), s, t \in \mathbb{R}$$

Cartesian equation

||

$$||x - x_1 y - y_1 z - z_1 x_2 - x_2 y_2 - y_2 z_2 - z_2 x_3 - x_3 y_3 - y_3 z_3 - z_3 x_1||$$

$$||=0||$$

$$||x - 1y + 1z - 32 - 14121||$$

$$||=0$$

$$(x-1)[-1-8]-(y+1)[z-4]+(z-3)$$

$$[4+1]=0-9(x-1)+2(y+1)+5(z-3)=0-9x+9+2y+2+5z-15=0-9x+2y+5z-4=0 \Rightarrow 9x-2y-5z+4=0$$

Question 6.

Find the parametric vector, non-parametric vector and Cartesian form of the equation of the plane passing through the point (3,6,-2), (-1,-2,6) and (6,4,-2).

Solution:

The required plane passing through the points

$$\vec{a} = 3\vec{i} + 6\vec{j} - 2\vec{k}, \vec{b} = -\vec{i} - 2\vec{j} + 6\vec{k} \text{ and } \vec{c} = 6\vec{i} - 4\vec{j} - 2\vec{k}$$

Parametric form of vector equation

$$\vec{r} = \vec{a} + s(\vec{b} - \vec{a}) + t(\vec{c} - \vec{a}) = (3\vec{i} + 6\vec{j} - 2\vec{k}) + s(-4\vec{i} - 8\vec{j} + 8\vec{k}) + t(3\vec{i} - 10\vec{j} + 4\vec{k})$$

Non-parametric form of vector equation

$$(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) = ||$$

|

$$||\vec{i} \vec{j} \vec{k}|| \begin{vmatrix} -4 & -8 & 8 \\ 3 & -10 & 4 \end{vmatrix}$$

|

$$= (-4)(-40) - (j)(0 - 24) + k(42 - 24) = 16i + 24j + 32k. (\vec{r} - \vec{a}) \cdot ((\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})) = 0 \Rightarrow (\vec{r} - \vec{a}) \cdot (16i + 24j + 32k) = 0 \Rightarrow (x-3)(16) + (y-6)(24) + (z+2)(32) = 0$$

$$(16x - 48) + (24y - 144) + (32z + 64) = 0 \Rightarrow 16x + 24y + 32z = 128 \Rightarrow 8[2x + 3y + 4z] = 128 \Rightarrow 2x + 3y + 4z = 16$$

Cartesian equation

$$(2x + 3y + 4z) \cdot (2i + 3j + 4k) = 16 \Rightarrow 2x + 3y + 4z = 16$$

Question 7.

Find the non-parametric form of vector equation, and Cartesian equations of the plane

$$\vec{r} = (6\vec{i} - \vec{j} + \vec{k}) + s(-\vec{i} + 2\vec{j} + \vec{k}) + t(-5\vec{i} - 4\vec{j} - 5\vec{k})$$

Solution:

$$\vec{a} = 6\vec{i} - \vec{j} + \vec{k}, \vec{b} = -\vec{i} + 2\vec{j} + \vec{k} \text{ and } \vec{c} = -5\vec{i} - 4\vec{j} - 5\vec{k}$$

$$\vec{b} \times \vec{c} = ||$$

|

$$||\vec{i} \vec{j} \vec{k}|| \begin{vmatrix} -1 & 2 & 1 \\ -5 & -4 & -5 \end{vmatrix}$$

|

$$= (-1)(-10) - j(5 + 5) + k(4 + 10) = 10i - 10j + 14k$$

Non-parametric form of vector equation

$$(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{c}) = 0 \Rightarrow (\vec{r} - \vec{a}) \cdot (10i - 10j + 14k) = 0 \Rightarrow (x-6)(10) - (y+1)(10) + (z+1)(14) = 0 \Rightarrow 10x - 10y + 14z = 40$$

Cartesian equation

$$(10x - 10y + 14z) \cdot (10i - 10j + 14k) = 40 \Rightarrow 10x - 10y + 14z = 40$$

Additional Problem

Question 1.

Find the vector and cartesian equations of the plane containing the line $x-22=y-23=z-13$ and parallel to the line $x+13=y-12=z+11$

Solution:

The required plane passes through the point A(2,2,1) and parallel to

$$\vec{u} = 2\vec{i} + 3\vec{j} + 3\vec{k} \text{ and } \vec{v} = 3\vec{i} + 2\vec{j} + \vec{k}$$

The required equation is $\vec{r} = \vec{a} + s\vec{u} + t\vec{v}$

$$\text{i.e., } x\vec{i} + y\vec{j} + z\vec{k} = (2\vec{i} + 2\vec{j} + \vec{k}) + s(2\vec{i} + 3\vec{j} + 3\vec{k}) + t(3\vec{i} + 2\vec{j} + \vec{k})$$

Cartesian form: The equation of the plane is

||

$$||x-x_1 \ y-y_1 \ z-z_1|| \cdot ||m_1 \ n_1 \ l_1 \ m_2 \ n_2 \ l_2||$$

$$||=0 \text{ i.e. } ||$$

$$||x-2 \ y-2 \ z-1|| \cdot ||2 \ 3 \ 3 \ 3 \ 2 \ 1||$$

$$||=0 \text{ i.e., } (x-2)[3-6]-(y-2)[2-9]+(z-1)[4-9]=0 \text{ i.e., } (x-2)(-3)+(y-2)(7)-5(z-1)=0-3x+6+7y-14-5z+5=0-3x+7y-5z-3=0 \text{ i.e. } 3x-7y+5z+3=0$$

Question 2.

Find the vector and cartesian equation of the plane passing through the point (1,3,2) and parallel to the lines $x+12=y+2-1=z+33$ and $x-21=y+12=z+22$.

Solution:

The required plane passes through the point $A=(1,3,2)$ and parallel to the vectors

$$\vec{u}=2\hat{i}+\hat{j}+3\hat{k} \text{ and } \vec{v}=2\hat{i}+2\hat{j}+2\hat{k}$$

The required equation is $\vec{r}=\vec{a}+s\vec{u}+t\vec{v}$

$$\text{i.e. } x\hat{i}+y\hat{j}+z\hat{k}=(1+2s+2t)\hat{i}+(3+s+2t)\hat{j}+(2+3s+2t)\hat{k}$$

Cartesian equation of the plane is ||

$$||x-x_1y-y_1z-z_1||m_1n_1l_1m_2n_2l_2||$$

$$||=0$$

So, the equation of the plane is ||

$$||x-1y-3z-22-13122||$$

$$||=0$$

$$\text{i.e. } (x-1)(-2-6)-(y-3)(4-3)+(z-2)(4+1)=0$$

$$-8(x-1)-1(y-3)+5(z-2)=0$$

$$-8x+8-y+3+5z-10=0$$

$$-8x-y+5z+1=0 \text{ i.e. } 8x+y-5z-1=0$$

Question 3.

Find the vector and cartesian equations of the plane passing through the point (-1,3,2) and perpendicular to the planes $x+2y+2z=5$ and $3x+y+2z=8$

Solution:

The normal vector to the given planes $x+2y+2z=5$ and $3x+y+2z=8$ are respectively $\vec{n}_1=\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{n}_2=3\hat{i}+\hat{j}+2\hat{k}$. These vectors are parallel to the required plane

The required plane passes through the point $A(-1,3,2)$ and parallel to the vectors

$$\vec{u}=\hat{i}+2\hat{j}+2\hat{k} \text{ and } \vec{v}=3\hat{i}+\hat{j}+2\hat{k}$$

The required equation is $\vec{r}=\vec{a}+s\vec{u}+t\vec{v}$

i.e.

$$\vec{r}=-\hat{i}+3\hat{j}+2\hat{k}+s(\hat{i}+2\hat{j}+2\hat{k})+t(3\hat{i}+\hat{j}+2\hat{k})$$

Cartesian form is ||

$$||x-x_1y-y_1z-z_1||m_1n_1l_1m_2n_2l_2||$$

$$||=0$$

So, the equation of the plane is ||

$$||x+1y-3z-2122312||$$

$$||=0$$

$$\text{i.e., } (x+1)(4-2)-(y-3)(2-6)+(z-2)(1-6)=0$$

$$2(x+1)+4(y-3)-5(z-2)=0$$

$$2x+2+4y-12-5z+10=0$$

$$2x+4y-5z=0$$

Question 4.

Find the vector and cartesian equations of the plane passing through the points $A(1,-2,3)$ and $B(-1,2,-1)$ and is parallel to the line $x-22=y+13=z-14$

Solution:

The vector equation of the plane is

$$\vec{r}=\vec{a}+s\vec{b}+t\vec{c} \text{ where } s \text{ and } t \text{ are scalars}$$

$$\therefore \vec{r}=(1-2s+2t)\hat{i}+(-2+3s+3t)\hat{j}+(3-s+t)\hat{k}$$

The cartesian equation of the plane is ||

$$||x-x_1y-y_1z-z_1||m_1n_1l_1m_2n_2l_2||$$

$$||=0$$

So, the equation of the plane is ||

$$||x-1y+2z-3-24-4234||$$

$$||=0$$

$$\text{i.e. } (x-1)[16+12]-(y+2)(-8+8)+(z-3)(-6-8)=0$$

$$28(x-1)-14(z-3)=0$$

$$28x-28-14z+42=0$$

$$28x-14z+14=0$$

$$(\div \text{ by } 14) \Rightarrow 2x-z+1=0$$

Question 5.

Find the vector and cartesian equations of the plane through the points (1,2,3) and (2,3,1) and perpendicular to the plane $3x-2y+4z-5=0$.

Solution:

The vector normal to the plane $3x-2y+4z-5=0$ is $3\mathbf{i}-2\mathbf{j}+4\mathbf{k}$

The required plane is parallel to the vector $3\mathbf{i}-2\mathbf{j}+4\mathbf{k}$

$\therefore$ Vector equation of the plane passing through two points and parallel to one vector is

$\mathbf{r} = (1-s)\mathbf{a} + t\mathbf{b} + t\mathbf{c}$ where s and t are scalars.

So, the vector equation of the required plane is

$$\mathbf{r} = (1-s)(\mathbf{i}+2\mathbf{j}+3\mathbf{k}) + s(2\mathbf{i}+3\mathbf{j}+\mathbf{k}) + t(3\mathbf{i}-2\mathbf{j}+4\mathbf{k})$$

The cartesian equation is ||

$$||x-x_1y-y_1z-z_1x_2-x_1y_2-y_1z_2-z_1||mn||$$

$$||=0$$

So, the equation of the plane is ||

$$||x-1y-2z-311-23-24||$$

$$||=0$$

$$\text{i.e. } (x-1)[4-4]-(y-2)[4+6]+(z-3)[-2-3]=0$$

$$-10(y-2)-5(z-3)=0$$

$$-10y+20-5z+15=0$$

$$-10y-5z+35=0$$

$$10y+5z-35=0$$

$$(\div \text{ by } 5) \Rightarrow 2y+z-1=0$$

Question 6.

Find the vector and cartesian equations of the plane containing the line $x-2=y-3=z-1$ and passing through the point $(-1,1,-1)$.

Solution:

The required plane passes through the points $A(-1,1,-1)$ and $B(2,2,1)$ and parallel to the vectors

$$\mathbf{c} = 2\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$$

The required equation is

$\mathbf{r} = (1-s)\mathbf{a} + t\mathbf{b} + t\mathbf{c}$ where s and t are scalars.

$$\mathbf{r} = (1-s)(-\mathbf{i}+\mathbf{j}-\mathbf{k}) + s(2\mathbf{i}+2\mathbf{j}+\mathbf{k}) + t(2\mathbf{i}+3\mathbf{j}-2\mathbf{k})$$

Cartesian equation of the plane is ||

$$||x-x_1y-y_1z-z_1x_2-x_1y_2-y_1z_2-z_1||mn||$$

$$||=0$$

So, the equation of the plane is ||

$$||x+1y-1z+131223-2||$$

$$||=0$$

$$\text{i.e. } (x+1)[-2-6]-(y-1)[-6-4]+(z+1)[9-2]=0$$

$$-8(x+1)+10(y-1)+7(z+1)=0$$

$$-8x-8+10y-10+7z+7=0$$

$$-8x+10y+7z-11=0$$

$$\text{i.e. } 8x-10y-7z+11=0$$

Question 7.

Find the vector and cartesian equations of the plane passing through the points with position vectors

$$3\vec{i}+2\vec{j}+\vec{k}, 2\vec{i}-2\vec{j}=\vec{k} \text{ and } 7\vec{i}+\vec{k}$$

Solution:

Vector equation of the plane passing through three given non-collinear points is

$\vec{r} = (1-s-t)\vec{a} + s\vec{b} + t\vec{c}$ where s and t are scalars $3(3\vec{i}+2\vec{j}+\vec{k}) + s(2\vec{i}-2\vec{j}-\vec{k}) + t(7\vec{i}+\vec{k})$ Cartesian equation of the plane is $||$

$$||x-x_1y-y_1z-z_1x_2-x_1y_2-y_1z_2-z_1x_3-x_1y_3-y_1z_3-z_1||$$

$$||=0 \Rightarrow ||$$

$$||x-3y-4z-2-1-6-34-4-1||$$

$$||=0 \text{ i.e. } (x-3)[6-12]-(y-4)[1+12]+(z-2)$$

$$[4+24]=0-6(x-3)-13(y-4)+28(z-2)=0-6x+18-13y+52+28z-56=0-6x-13y+28z+14=0 \text{ i.e. } 6x+13y-28z-14=0$$

Question 8.

Derive the equation of the plane in the intercept form.

Solution:

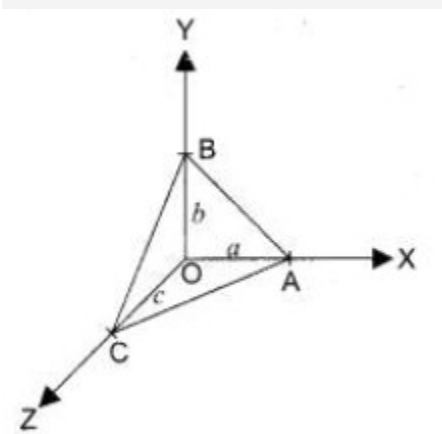
Let the required plane makes intercepts on X, Y, Z -axes respectively as a, b and c .

$$\text{i.e. } A=(a, 0, 0)$$

$$B=(0, b, 0)$$

$$C=(0, 0, c)$$

The equation of the plane passing through three non-collinear points A, B and C is



$\vec{r} = (1-s-t)\vec{a} + s\vec{b} + t\vec{c}$. Here $\vec{a} = a\vec{i}, \vec{b} = b\vec{j}$ and $\vec{c} = c\vec{k}$. $\vec{r} = (1-s-t)a\vec{i} + sb\vec{j} + tc\vec{k}$ Cartesian equation of the plane is $||$

$$||x-x_1y-y_1z-z_1x_2-x_1y_2-y_1z_2-z_1x_3-x_1y_3-y_1z_3-z_1||$$

$$||=0 \Rightarrow ||$$

$$||x-ay-0z-0-ab0-a0c||$$

$$||=0(x-a)(bc-0)-y(-ac-0)+z(ab)=0 \text{ i.e. } xbc-abc+yac+zab=0 \text{ i.e. } xbc+yac+zab=abc \text{ (+ by abc) } \Rightarrow xa+yb+zc=1$$

Question 9.

Find the cartesian form of the following plane:

$$(s-2t)\vec{i} + (3-t)\vec{j} + (2s+t)\vec{k}$$

Solution:

Now

$$x\vec{i} + y\vec{j} + z\vec{k}$$

$$(s-2t)\vec{i} + (3-t)\vec{j} + (2s+t)\vec{k} = (s-2t)\vec{i} + (-t)\vec{j} + (2s+t)\vec{k}$$

Equating $\vec{i}, \vec{j}, \vec{k}$ components and eliminating s and t we get,

$$||$$

$$||x-0y-3z-0s02s-2t-t||$$

$$||=0; \text{ i.e., } ||$$

$$||xy-3z102-2-11||$$

$$||=0x(0+2)-(y-3)(1+4)+z(-1)=02x-5(y-3)-z=02x-5y+15-z=02x-5y-z+15=0$$

which is the cartesian equation of the plane.

Ex 6.8

Question 1.

Show that the straight lines $\vec{r} = (5\hat{i} + 7\hat{j} - 3\hat{k}) + s(4\hat{i} + 4\hat{j} - 5\hat{k})$ and $\vec{r} = (8\hat{i} + 4\hat{j} + 5\hat{k}) + t(7\hat{i} + \hat{j} + 3\hat{k})$ are coplanar. Find the vector equation of the plane in which they lie.

Solution:

Let $\vec{a} = 5\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{b} = 4\hat{i} + 4\hat{j} - 5\hat{k}$

$\vec{c} = 8\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{d} = 7\hat{i} + \hat{j} + 3\hat{k}$

We know that given two lines are coplanar if

$$(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = 0$$

|

$$||\vec{b} \times \vec{d}|| = ||17\hat{i} - 47\hat{j} - 24\hat{k}||$$

|

$$= (17)^2 + (-47)^2 + (-24)^2 = 338$$

$$(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = (3\hat{i} - 3\hat{j} + 8\hat{k}) \cdot (17\hat{i} - 47\hat{j} - 24\hat{k}) = 51 - 141 - 192 = 0$$

∴ The two given lines are coplanar so, the non-parametric vector equation is

$$(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = 0 \Rightarrow (\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = 0 \Rightarrow (\vec{r} - \vec{a}) \cdot (17\hat{i} - 47\hat{j} - 24\hat{k}) = 0$$

$$(17i - 47j - 24k) \cdot (17i - 47j - 24k) = 85 - 329 + 72 \Rightarrow r \cdot (17i - 47j - 24k) = -172$$

Question 2.

Show that lines $x - 21 = y - 31 = z - 43$ and $x - 1 - 3 = y - 42 = z - 51$ are coplanar. Also, find the plane containing these lines.

Solution:

From the lines we have, $(x_1, y_1, z_1) = (2, 3, 4)$ and $(x_2, y_2, z_2) = (1, 4, 5)$

$(b_1, b_2, b_3) = (1, 1, 3)$ and $(d_1, d_2, d_3) = (-3, 2, 1)$

Condition for coplanarity

||

$$||x_2 - x_1, y_2 - y_1, z_2 - z_1|| = ||-1, 1, 1||$$

$$= 0$$

$$= -111113 - 321$$

$$= -(1-6) - 1(1+9) + 1(2+3) = 5 - 10 + 5 = 0$$

∴ The given two lines are coplanar.

Cartesian form of equation of the plane containing the two given coplanar lines.

||

$$||x - x_1, y - y_1, z - z_1|| = ||x - 2, y - 3, z - 4||$$

$$= 0$$

$$= 0(x-2)[1-6] - (y-3)[1+9] + (z-4)[2+3]$$

$$= 0(x-2)[-5] - (y-3)[10] + (z-4)[5]$$

$$= 0 - 5(x-2) - 10(y-3) + 5(z-4) = 0 - 5x + 10 - 10y + 30 + 5z - 20 = 0 - 5x - 10y + 5z + 20 = 0 \Rightarrow x + 2y - z - 4 = 0$$

Question 3.

If the straight lines $x - 11 = y - 22 = z - 3m_2$ and $x - 31 = y - 2m_2 = z - 12$ are coplanar, find the distinct real values of m .

Solution:

From the lines we have, $(x_1, y_1, z_1) = (1, 2, 3)$ and $(x_2, y_2, z_2) = (3, 2, 1)$

$(b_1, b_2, b_3) = (1, 2, m_2)$ and $(d_1, d_2, d_3) = (1, m_2, 2)$

Condition for coplanarity

||

$$||x_2 - x_1, y_2 - y_1, z_2 - z_1|| = ||2, 0, -2||$$

$$= 0$$

|

$$= 20 - 212m_2 + m_2^2$$

|

$$||=0 \Rightarrow 2[4-m_4]-2[m_2-2]=0 \quad (+ \text{ by } 2) \Rightarrow 4-m_4-m_2+2=0 \Rightarrow m_4+m_2-6=0 \quad (m_2+3)(m_2-2)=0 \Rightarrow m_2+3=0 \Rightarrow m_2=-3 \Rightarrow m_2=2 \quad (\text{not possible}) \Rightarrow m=\pm\sqrt{2}$$

Question 4.

If the straight lines $x-12=y+1=z$ and $x+15=y+12=z$: are coplanar, find λ and equation of the planes containing these two lines.

Solution:

From the lines we have, $(x_1, y_1, z_1)=(1, -1, 0)$ and $(x_2, y_2, z_2)=(-1, -1, 0)$

$(b_1, b_2, b_3)=(2, \lambda, 2)$ and $(d_1, d_2, d_3)=(5, 2, \lambda)$

Condition for coplanarity

Try Prep Courses

$$-2(\lambda-4)=0 \Rightarrow \lambda=4 \Rightarrow \lambda=\pm 2 \quad (\text{i) If } \lambda=2$$

$$||x-x_1y-y_1z-z_1b_1b_2b_3d_1d_2d_3||$$

$$||=0||$$

$$||x-1y+1z-222522||$$

$$||=0 \Rightarrow (x-1)[0]-(y+1)[4-10]+z[4-10]=0 \Rightarrow 6(y+1)-6(z)=0 \Rightarrow 6y+6-6z=0 \quad (+ \text{ by } 6) \Rightarrow (y-z+1)=0$$

$$||x-x_1y-y_1z-z_1b_1b_2b_3d_1d_2d_3||$$

$$||=0||$$

$$||x-1y+1z-2252-2||$$

$$||=0 \Rightarrow (x-1)[0]-(y+1)[-4-10]+z[4+10]=0 \Rightarrow 14(y+1)+14z=0 \Rightarrow 14y+14+14z=0 \quad (+ \text{ by } 14) \Rightarrow y+z+1=0$$

Additional Problems

Question 1.

Show that the straight lines.

$$\vec{r}=(4i+3j-k)+\lambda(3i-j) \quad \text{and} \quad \vec{r}=(4i-k)+\mu(2i+3k)$$

are coplanar. Find the vector equation of the plane in which they lie.

Solution:

Given $\vec{a}=4i+3j-k$; and $\vec{b}=3i-j$

$$\vec{c}=4i-k; \text{ and } \vec{d}=2i+3k$$

Conditions for coplanar

$$(\vec{c}-\vec{a}) \cdot (\vec{b} \times \vec{d})=0 \Rightarrow \vec{c}-\vec{a}=(4i-k)-(4i+3j-k)=3i-j \Rightarrow \vec{b} \times \vec{d}=||$$

$$|$$

$$||ijk|-10203||$$

$$|$$

$$||=(3i-j) \cdot (3i-j) \cdot (2i+3k)=-3i-9j+2k \Rightarrow (\vec{c}-\vec{a}) \cdot (\vec{b} \times \vec{d})=(-9+9+0)=0.$$

Given two lines are coplanar.

Vector equation of the plane:

$$\vec{r}=\vec{a}+\vec{t}\vec{b}+\vec{s}\vec{d} \Rightarrow \vec{r}=(4i+3j-k)+\vec{t}(3i-j)+\vec{s}(2i+3k) \Rightarrow \vec{r}=\vec{c}+\vec{t}\vec{b}+\vec{s}\vec{d} \Rightarrow \vec{r}=(4i-k)+\vec{t}(3i-j)+\vec{s}(2i+3k)$$

Question 2.

If the straight lines $x-11=y-1=z-11$ and $x-2=y-4=z-23$ are coplanar. Find λ .

Solution:

From the lines we have, $(x_1, y_1, z_1)=(1, 1, 1)$ and $(b_1, b_2, b_3)=(1, \lambda, 1)$

$(x_2, y_2, z_2)=(0, 4, 2)$ and $(d_1, d_2, d_3)=(2, \lambda, 3)$

Condition for coplanarity

$$||$$

$$||x_2-x_1y_2-y_1z_2-z_1b_1b_2b_3d_1d_2d_3||$$

$$||=0||$$

$$||-1311\lambda12\lambda3||$$

$$||=0 \Rightarrow -1(3\lambda-\lambda)-3(3-2)+1(\lambda-2\lambda)=0 \Rightarrow -2\lambda-3-\lambda=0 \Rightarrow -3\lambda=3 \Rightarrow \lambda=-1$$

Question 3.

If the lines $x-21=y-31=z-4-k$ and $x-1k=y-42=z-51$ are coplanar, then find the value of k . **Solution:**

From the lines we have, $(x_1, y_1, z_1) = (2, 3, 4)$ and $(b_1, b_2, b_3) = (1, 1, -1)$ $(x_2, y_2, z_2) = (1, 4, 5)$ and $(d_1, d_2, d_3) = (k, 2, 1)$

Condition for coplanarity

||

$$||x_2 - x_1 y_2 - y_1 z_2 - z_1 b_1 b_2 b_3 d_1 d_2 d_3||$$

$$||=0||$$

$$||-11111-kk21||$$

$$||=0$$

$$-1(1+2k)-1(1+k_2)+1(2-k)=0$$

$$-1-2k-1-k_2+2-k=0-k_2-3k=0k_2+3k=0k(k+3)=0k=0 \text{ or } k=-3$$

Question 4.

Show that the lines $x+3=z-5$ and $x+1=y-2=z-5$ are coplanar, Also find the equation of the plane containing these two lines.

Solution:

From the lines we have, $(x_1, y_1, z_1) = (-3, 1, 5)$ and $(b_1, b_2, b_3) = (-3, 1, 5)$

$(x_2, y_2, z_2) = (-1, 2, 5)$ and $(d_1, d_2, d_3) = (-1, 2, 5)$

Condition for coplanarity

||

$$||x_2 - x_1 y_2 - y_1 z_2 - z_1 b_1 b_2 b_3 d_1 d_2 d_3||$$

$$||=0||$$

$$||210-315-125||$$

$$||=2(5-10)-1(-15+5)+0(-6+1)=-10+10=0$$

Given two lines are coplanar

||

$$||x - x_1 y - y_1 z - z_1 b_1 b_2 b_3 d_1 d_2 d_3||$$

$$||=0||$$

$$||x+3y-1z-5-315-125||$$

$$||=0(x+3)[5-10]-(y-1)[-15+5]+(z-5)[-6+1]=05(x+3)+10(y-1)-5(z-5)=0(+ \text{ by } 5) \Rightarrow (x+3)-2(y-1)+(z-5)=0x+3-2y+2+z-5=0x-2y+z=0 \text{ or } ||$$

$$||x - x_2 y - y_2 z - z_2 b_1 b_2 b_3 d_1 d_2 d_3||$$

$$||=0||$$

$$||x+1y-2z-5-315-125||$$

$$||=0(x+1)(5-10)-(y-2)(-15+5)+(z-5)(-6+1)=0-5(x+1)+10(y-2)-5(z-5)=0(x+1)-2(y-2)+(z-5)=0x+1-2y+4+z-5=0x-2y+z=0$$

Ex 6.9

Question 1.

Find the equation of the plane passing through the line of intersection of the planes $\vec{r} \cdot (2\hat{i} - 7\hat{j} + 4\hat{k}) = 3$ and $3x - 5y + 4z + 11 = 0$, and the point $(-2, 1, 3)$.

Solution:

Given planes are

$$\vec{r} \cdot (2\hat{i} - 7\hat{j} + 4\hat{k}) = 3$$

$$2x - 7y + 4z - 3 = 0 \text{ and } 3x - 5y + 4z + 11 = 0$$

Equation of a plane which passes through the line of intersection of the planes

$$(2x - 7y + 4z - 3) + \lambda(3x - 5y + 4z + 11) = 0$$

This passes through the point $(-2, 1, 3)$.

$$(1) \Rightarrow (-4 - 7 + 12 - 3) + \lambda(-6 - 5 + 12 + 11) = 0 - 2 + \lambda(12) = 0 \Rightarrow 12\lambda = 2\lambda = 212 \Rightarrow \lambda = 16$$

The required equation is

$$(1) \Rightarrow (2x - 7y + 4z - 3) + 16(3x - 5y + 4z + 11) = 012x - 42y + 24z - 18 + 3x - 5y + 4z + 11 = 015x - 47y + 28z - 7 = 0$$

Question 2.

Find the equation of the plane passing through the line of intersection of the planes $x + 2y + 3z = 2$ and $x - y$

$+z+11=3$, and at a distance $\frac{20}{3}$ from the point $(3,1,-1)$.

Solution:

Equation of a plane which passes through the line of intersection of the plane

$$(x+2y+3z-2)+\lambda(x-y+z-3)=0$$

$$(1+\lambda)x + (2-\lambda)y + (3+\lambda)z + (-2-3\lambda) = 0 \dots\dots\dots (1)$$

Given
$$\left| \frac{3(\lambda+1)+1(2-\lambda)-1(3+\lambda)-2-3\lambda}{\sqrt{(\lambda+1)^2+(2-\lambda)^2+(\lambda+3)^2}} \right| = 0$$

$$\left| \frac{3\cancel{\lambda} + \cancel{\lambda} + \cancel{\lambda} - \lambda - \cancel{\lambda} - \lambda - \cancel{\lambda} - 3\cancel{\lambda}}{\sqrt{\lambda^2+1+2\lambda+4+\lambda^2-4\lambda+\lambda^2+9+6\lambda}} \right| = \frac{2}{\sqrt{3}}$$

$$\left| \frac{-2\lambda}{\sqrt{3\lambda^2+4\lambda+14}} \right| = \frac{2}{\sqrt{3}}$$

Squaring on both sides
$$\frac{\cancel{A}\lambda^2}{3\lambda^2+4\lambda+14} = \frac{\cancel{A}}{3}$$

$$3\cancel{\lambda}^2 = 3\cancel{\lambda}^2 + 4\lambda + 14$$

$$4\lambda = -14$$

$$\lambda = \frac{-7}{2}$$

Substituting in (1)

$$(1) \Rightarrow \left(1 - \frac{7}{2}\right)x + \left(2 + \frac{7}{2}\right)y + \left(3 - \frac{7}{2}\right)z - 2 - 3\left(\frac{-7}{2}\right) = 0$$

$$\left(\frac{-5}{2}\right)x + \left(\frac{11}{2}\right)y + \left(\frac{-1}{2}\right)z + \frac{17}{2} = 0$$

$$-5x + 11y - z + 17 = 0$$

(or)
$$5x - 11y + z - 17 = 0$$

Question 3.

Find the angle between the line $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + t(\hat{i} + 2\hat{j} - 2\hat{k})$ and the plane $\vec{r} \cdot (6\hat{i} + 3\hat{j} + 2\hat{k}) = 8$.

Solution:

Angle between the line and a plane

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|} \quad \vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}, \vec{n} = 6\hat{i} + 3\hat{j} + 2\hat{k} \quad \vec{b} \cdot \vec{n} = 6 + 6 - 4 = 8 \quad |\vec{b}| = \sqrt{1+4+4} = \sqrt{9} = 3 \quad |\vec{n}| = \sqrt{36+9+4} = \sqrt{49} = 7$$

$$(1) \Rightarrow \sin \theta = \frac{8}{(3)(7)} \Rightarrow \theta = \sin^{-1}\left(\frac{8}{21}\right)$$

Question 4.

Find the angle between the planes $\vec{r} \cdot (\hat{i} + \hat{j} - 2\hat{k}) = 3$ and $2x - 2y + z = 2$.

Solution:

Angle between given two planes

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} \quad \vec{n}_1 = \hat{i} + \hat{j} - 2\hat{k}, \vec{n}_2 = 2\hat{i} - 2\hat{j} + \hat{k} \quad \vec{n}_1 \cdot \vec{n}_2 = 2 - 2 - 2 = -2 \quad |\vec{n}_1| = \sqrt{1+1+4} = \sqrt{6} \quad |\vec{n}_2| = \sqrt{4+4+1} = 3 \quad \cos \theta = \frac{-2}{3\sqrt{6}} = -\frac{2}{3\sqrt{6}} \Rightarrow \theta = \cos^{-1}\left(\frac{2}{23\sqrt{6}}\right)$$

Question 5.

Find the equation of the plane which passes through the point (3,4,-1) and is parallel to the plane $2x-3y+5z+7=0$. Also, find the distance between the two planes.

Solution:

The required equation parallel to the plane

$$2x-3y+5z+7=0 \Rightarrow 2x-3y+5z+\lambda=0$$

This passes through (3,4,-1)

$$(2) \Rightarrow 2(3)-3(4)+5(-1)+\lambda=0$$

$$6-12-5+1=0$$

$$\lambda=11$$

$$(2) \Rightarrow \text{The required equation is } 2x-3y+5z+11=0$$

$\therefore$ Now, distance between the above parallel lines (1) and (3)

$$a=2, b=-3, c=5, d_1=7, d_2=11 \text{ Distance} = \frac{|d_2-d_1|}{\sqrt{a^2+b^2+c^2}} = \frac{|11-7|}{\sqrt{(2)^2+(-3)^2+(5)^2}}$$

$$= \frac{4}{\sqrt{4+9+25}} = \frac{4}{\sqrt{38}}$$

$$= \frac{4}{\sqrt{38}}$$

Question 6.

Find the length of the perpendicular from the point (1,-2,3) to the plane $x-y+z=5$.

Solution:

Perpendicular length from the point (x_1, y_1, z_1) to the plane $ax+by+cz+d=0$ is

$$\frac{|ax_1+by_1+cz_1+d|}{\sqrt{a^2+b^2+c^2}}$$

Given point (1,-2,3) and the plane $x-y+z=5$

$$\therefore \text{Length of the perpendicular} = \frac{|1-(-2)+3-5|}{\sqrt{(1)^2+(-1)^2+(1)^2}} = \frac{|1+2+3-5|}{\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ units}$$

Question 7.

Find the point of intersection of the line $x-1=y=z+1$ with the plane $2x-y+2z=2$. Also, find the angle between the line and the plane.

Solution:

Any point on the line $x-1=y=z+1$ is

$$x-1=y=z+1=\lambda, \text{ (say) } (\lambda+1, \lambda, \lambda-1)$$

This passes through the plane $2x-y+2z=2$

$$2(\lambda+1)-\lambda+2(\lambda-1)=2 \Rightarrow 2\lambda+2-\lambda+2\lambda-2=2 \Rightarrow 3\lambda=2 \Rightarrow \lambda=\frac{2}{3}$$

$\therefore$ The required point of intersection is $(\frac{5}{3}, \frac{2}{3}, -\frac{1}{3})$

Angle between the line and the plane is

$$\sin \theta = \frac{|\vec{a} \cdot \vec{n}|}{|\vec{a}| |\vec{n}|} \text{ where } \vec{a} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k}) \text{ and } \vec{n} = 2\hat{i} - \hat{j} + 2\hat{k} \Rightarrow \vec{a} \cdot \vec{n} = \frac{1}{\sqrt{3}}(1-1+2) = \frac{2}{\sqrt{3}}$$

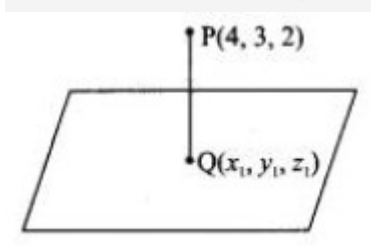
$$|\vec{a}| = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}, |\vec{n}| = \sqrt{4+1+4} = \sqrt{9} = 3 \Rightarrow \sin \theta = \frac{\frac{2}{\sqrt{3}}}{\frac{1}{\sqrt{3}} \cdot 3} = \frac{2}{3} \Rightarrow \theta = \sin^{-1}\left(\frac{2}{3}\right)$$

Question 8.

Find the coordinates of the foot of the perpendicular and length of the perpendicular from the point (4,3,2) to the plane $x+2y+3z=2$

Solution:

Direction of the normal plane (1,2,3)



d.c.s of the PQ is $x_1-4=y_1-3=z_1-2=k$

$$x_1=k+4, y_1=k+3, z_1=k+2$$

This passes through the plane $x+2y+3z=2$

$$k+4+2(k+3)+3(k+2)=2$$

$$k+4+2k+6+3k+6=2 \Rightarrow 6k+16=2 \Rightarrow 6k=-14 \Rightarrow k=-\frac{7}{3}$$

$$14k = 2 - 16 \Rightarrow 14k = -14$$

$$k = -1$$

☒ The coordinate of the foot of the perpendicular is (3,1,-1)

☒ Length of the perpendicular to the plane is

$$= ||4 + 2(3) + 3(2) - 2\sqrt{(1)^2 + (2)^2 + (3)^2}|| = ||4 + 6 + 6 - 2\sqrt{1+4+9}|| = 14\sqrt{14} = \sqrt{14} \text{ units}$$

$$(1 + \lambda)x + (2 - \lambda)y + (3 + \lambda)z + (-2 - 3\lambda) = 0 \dots\dots\dots (1)$$

Given
$$\left| \frac{3(\lambda + 1) + 1(2 - \lambda) - 1(3 + \lambda) - 2 - 3\lambda}{\sqrt{(\lambda + 1)^2 + (2 - \lambda)^2 + (\lambda + 3)^2}} \right| = 0$$

$$\left| \frac{\cancel{3\lambda} + \cancel{\lambda} + \cancel{2} - \lambda - \cancel{\lambda} - \cancel{2} - \cancel{3\lambda}}{\sqrt{\lambda^2 + 1 + 2\lambda + 4 + \lambda^2 - 4\lambda + \lambda^2 + 9 + 6\lambda}} \right| = \frac{2}{\sqrt{3}}$$

$$\left| \frac{-2\lambda}{\sqrt{3\lambda^2 + 4\lambda + 14}} \right| = \frac{2}{\sqrt{3}}$$

Squaring on both sides

$$\frac{\cancel{4}\lambda^2}{3\lambda^2 + 4\lambda + 14} = \frac{\cancel{4}}{3}$$

$$\cancel{3\lambda^2} = \cancel{3\lambda^2} + 4\lambda + 14$$

$$4\lambda = -14$$

$$\lambda = \frac{-7}{2}$$

Substituting in (1)

$$(1) \Rightarrow \left(1 - \frac{7}{2}\right)x + \left(2 + \frac{7}{2}\right)y + \left(3 - \frac{7}{2}\right)z - 2 - 3\left(\frac{-7}{2}\right) = 0$$

$$\left(\frac{-5}{2}\right)x + \left(\frac{11}{2}\right)y + \left(\frac{-1}{2}\right)z + \frac{17}{2} = 0$$

$$-5x + 11y - z + 17 = 0$$

(or)

$$5x - 11y + z - 17 = 0$$

Additional Problems

Question 1.

Find the point of intersection of the line passing through the two points (1,1,-1);(-1,0,1) and the xy-plane.

Solution:

The equation of the line passing through (1,1,-1) and (-1,0,1) is

$$x-12=y-11=z+1-2$$

It meets the xy-plane i.e. z=0

$$\therefore x-12=y-11=1-2 \Rightarrow x=0, y=12$$

The required point is (0,12,0).

Question 2.

Find the co-ordinates of the point where the line $\vec{r} = (\hat{i} + 2\hat{j} - 3\hat{k}) + t(2\hat{i} - 3\hat{j} + 4\hat{k})$ meets the plane $\vec{r} \cdot (2\hat{i} + 4\hat{j} - \hat{k}) = 3$.

Solution:

The equation of the straight line in the cartesian form is

$$x-12=y-2-3=z+54=\lambda \text{ (say)}$$

$\therefore$ Any point on this line is of the form $(2\lambda+1, -3\lambda, +2, 4\lambda-5)$

The cartesian equation of the plane is $2x+4y-z=3$

But the required point lies on this plane.

$$\therefore 2(2\lambda+1)+4(-3\lambda+2)-(4\lambda-5)-3=0 \Rightarrow \lambda=1$$

$\therefore$ The required point is $(3, -1, -1)$.

Question 3.

Find the point of intersection of the line $\vec{r} = (\hat{j} - \hat{k}) + s(2\hat{i} - \hat{j} + \hat{k})$ and xz -plane.

Solution:

The given point is $(0, 1, -1)$; parallel vector is $2\hat{i} - \hat{j} + \hat{k}$.

The equation of the line passing through the point $(0, 1, -1)$ and parallel to the vector $2\hat{i} - \hat{j} + \hat{k}$ is

$$x-0/2=y-1/-1=z+1/1$$

It meets the xz -plane i.e. $y=0$

$$\Rightarrow x/2=1/-1=z+1/1 \Rightarrow x/2=1 \Rightarrow x=2, z+1=1 \Rightarrow z=0$$

$\therefore$ The required point is $(2, 0, 0)$

Question 4.

Find the meeting point of the line $\vec{r} = (2\hat{i} + \hat{j} - 3\hat{k}) + t(2\hat{i} - \hat{j} - \hat{k})$ and the plane $x-2y+3z+7=0$.

Solution:

The equation of the straight line in cartesian form is $x-22=y-1-1=z+3-1=\lambda$ (say)

Any point on the line is $(2\lambda+2, -\lambda+1, \lambda-3)$

The point lies on the plane $x-2y+3z+7=0$

$$\Rightarrow 2\lambda+2-2(-\lambda+1)+3(\lambda-3)+7=0 \Rightarrow 2\lambda+2+2\lambda-2-3\lambda-9+7=0 \Rightarrow \lambda-2=0 \Rightarrow \lambda=2$$

So $2\lambda+2=6$; $-\lambda+1=-1$; $\lambda-3=-5$ The required point is $(6, -1, -5)$

Question 5.

Show that the following planes are at right angles:

$$\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 15 \text{ and } \vec{r} \cdot (\hat{i} - \hat{j} - 3\hat{k}) = 3$$

Solution:

The normal vectors are

$$\vec{n}_1 = 2\hat{i} - \hat{j} + \hat{k} \text{ and } \vec{n}_2 = \hat{i} - \hat{j} - 3\hat{k}$$

$$\vec{n}_1 \cdot \vec{n}_2 = (2)(1) + (-1)(-1) + (1)(-3) = 2 + 1 - 3 = 0$$

$\Rightarrow \vec{n}_1 \perp \vec{n}_2$ to $\vec{n}_2$, i.e., the normals to the planes are at right angles. So, the planes are at right angles.

Question 6.

The planes $2(\hat{i} + \hat{j} - 3\hat{k}) = 10$ and $5(\hat{i} + 3\hat{j} + \hat{k}) = 5$ are perpendicular. Find λ .

Solution:

Since the planes are perpendicular, the angle between the normals $= 90^\circ$.

The normals are $\vec{n}_1 = 2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{n}_2 = \lambda\hat{i} + 3\hat{j} + \hat{k}$ $\vec{n}_1 \cdot \vec{n}_2 = 0$ [$\because \angle = 90^\circ$]

$$\Rightarrow (2)(\lambda) + (1)(3) + (-3)(1) = 0 \Rightarrow 2\lambda + 3 - 3 = 0$$

$$5\lambda - 3 = 0 \Rightarrow 5\lambda = 3 \Rightarrow \lambda = 3/5$$

Question 7.

Find the angle between the line $x-23=y+1-1=z-3-2$ and the plane $3x+4y+z=0$.

Solution:

The angle between the line $\vec{r} = \vec{a} + t\vec{b}$ and the plane $\vec{r} \cdot \vec{n} = p$ is given by the formula

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|} \text{ So, } \sin \theta = \frac{|(3\hat{i} - \hat{j} - 2\hat{k}) \cdot (3\hat{i} + 4\hat{j} + \hat{k})|}{|(3\hat{i} - \hat{j} - 2\hat{k})| |(3\hat{i} + 4\hat{j} + \hat{k})|}$$

$$= \frac{|9 - 4 - 2|}{\sqrt{9+1+4} \sqrt{9+16+1}} = \frac{3}{\sqrt{14} \sqrt{26}} = \frac{3}{\sqrt{364}} = \frac{3}{\sqrt{4 \times 91}} = \frac{3}{2\sqrt{91}}$$

$$\text{So, } \theta = \sin^{-1} \left(\frac{3}{2\sqrt{91}} \right)$$

Question 8.

Find the angle between the line $\vec{r} = \hat{i} + \hat{j} + 3\hat{k} + \lambda(2\hat{i} - \hat{j} - \hat{k})$ and the plane $\vec{r} \cdot (\hat{i} + \hat{j}) = 1$.

Solution:

Here $\vec{b} = 2\hat{i} - \hat{j} - \hat{k}$ and $\vec{n} = \hat{i} + \hat{j}$

The angle between the line and the plane is given by

$$\sin \theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$$

$$\vec{b} \cdot \vec{n} = (2\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} + \hat{j}) = 2 + 1 = 3, |\vec{b}| = \sqrt{4+1+1} = \sqrt{6}, |\vec{n}| = \sqrt{1+1} = \sqrt{2} \therefore \sin \theta = \frac{3}{\sqrt{6}\sqrt{2}} = \frac{3}{\sqrt{12}} = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2} \therefore \theta = \frac{\pi}{3}$$

Ex 6.10

Choose the correct or the most suitable answer from the given four alternatives:

Question 1.

If $\vec{a}$ and $\vec{b}$ are parallel vector, then $[\vec{a}, \vec{c}, \vec{b}]$ is equal to

- (a) 2
- (b) -1
- (c) 1
- (d) 0

Solution:

(d) 0

Hint:

$\vec{a}$ and $\vec{b}$ are parallel vectors, so $\vec{a} \times \vec{b} = 0$

then $[\vec{a}, \vec{c}, \vec{b}] = [\vec{a} \times \vec{b}, \vec{c}] = [0, \vec{c}] = 0$

Question 2.

If a vector $\vec{\alpha}$ lies in the plane of $\vec{\beta}$ and $\vec{\gamma}$, then .

- (a) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 1$
- (b) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = -1$
- (c) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$
- (d) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 2$

Solution:

(c) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$

Hint:

Since $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}]$ are lie in the same plane

so $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$

Question 3.

If $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$, then the value of $[\vec{a}, \vec{b}, \vec{c}]$ is

- (a) $|\vec{a}| |\vec{b}| |\vec{c}|$
- (b) $\frac{1}{3} |\vec{a}| |\vec{b}| |\vec{c}|$
- (c) 1
- (d) -1

Solution:

(a) $|\vec{a}| |\vec{b}| |\vec{c}|$

Hint:

$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$

$\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors

$[\vec{a}, \vec{b}, \vec{c}] = |\vec{a}| |\vec{b}| |\vec{c}|$ is volume of cuboid where $\vec{a}, \vec{b}, \vec{c}$ are coterminal edges.

Question 4.

If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}$ is perpendicular to $\vec{b}$, and is parallel to $\vec{c}$ then $\vec{a} \times (\vec{b} \times \vec{c})$ is equal to .

- (a) $\vec{a}$
- (b) $\vec{b}$
- (c) $\vec{c}$
- (d) $\vec{0}$

Solution:

(b) $\vec{b}$

Hint:

$\vec{a}$ is perpendicular to $\vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$

$\vec{a}$ is parallel to $\vec{c} \Rightarrow \vec{a} \times \vec{c} = 0$

then $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

$\Rightarrow 0 - 0 = -\vec{0} = \vec{0}$

Question 5.

If $[\vec{a}, \vec{b}, \vec{c}] = 1$, then the value of $-(\vec{b} \times \vec{c}) \cdot \vec{a} + (\vec{c} \times \vec{a}) \cdot \vec{b} - (\vec{a} \times \vec{b}) \cdot \vec{c}$ is

- (a) 1
- (b) -1
- (c) 2
- (d) 3

Solution:

- (a) 1

Hint:

$[\vec{a}, \vec{b}, \vec{c}] = 1$ (Given) $-(\vec{b} \times \vec{c}) \cdot \vec{a} + (\vec{c} \times \vec{a}) \cdot \vec{b} - (\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a}, \vec{b}, \vec{c}] - [\vec{a}, \vec{b}, \vec{c}] + [\vec{a}, \vec{b}, \vec{c}] - [\vec{a}, \vec{b}, \vec{c}] = 1 + 1 - 1 = 1$

Question 6.

The volume of the parallelepiped with its edges represented by the vectors $\hat{i} + \hat{j}, \hat{i} + 2\hat{j}, \hat{i} + \hat{j} + \pi\hat{k}$ is

- (a) π^2
- (b) π^3
- (c) π
- (d) π^4

Solution:

- (c) π

Hint:

Volume $[\vec{a}, \vec{b}, \vec{c}] = \left| \begin{vmatrix} 1 & 1 & \pi \\ 1 & 2 & 0 \\ 1 & 1 & \pi \end{vmatrix} \right|$

$= \pi(2-1) = \pi$ cubic units

Question 7.

If $\vec{a}$ and $\vec{b}$ are unit vectors such that $[\vec{a}, \vec{b}, \vec{a} \times \vec{b}] = \pi/4$ then the angle between $\vec{a}$ and $\vec{b}$ is

- (a) $\pi/6$
- (b) $\pi/4$
- (c) $\pi/3$
- (d) $\pi/2$

Solution:

- (a) $\pi/6$

Hint:

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$$\begin{aligned} [\vec{a}, \vec{b}, \vec{a} \times \vec{b}] &= \frac{1}{4} \\ (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) &= \frac{1}{4} \\ (\vec{a} \times \vec{b})^2 &= \frac{1}{4} \\ |\vec{a} \times \vec{b}|^2 &= \frac{1}{4} \\ |\vec{a} \times \vec{b}| &= \frac{1}{2} \end{aligned} \quad \left| \quad \begin{aligned} |\vec{a}| |\vec{b}| \sin \theta &= \frac{1}{2} \\ (1)(1) \sin \theta &= \frac{1}{2} \\ \sin \theta &= \frac{1}{2} \\ \theta &= \sin^{-1} \left(\frac{1}{2} \right) \\ \theta &= \frac{\pi}{6} \end{aligned} \right.$$

Question 8.

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} + \hat{j}, \vec{c} = \hat{i}$ and $(\vec{a} \times \vec{b}) \times \vec{c} = \lambda \vec{a} + \mu \vec{b}$, then the value of $\lambda + \mu$ is

- (a) 0
- (b) 1
- (c) 6
- (d) 3

Solution:

(a) 0

Hint:

$$\text{But } (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} \equiv \vec{b} - \vec{a} \Rightarrow (\vec{a} \times \vec{b}) \times \vec{c} = \vec{\lambda}\vec{a} + \vec{\mu}\vec{b} \Rightarrow \vec{a} = \vec{\lambda}\vec{a} + \vec{\mu}\vec{b} \Rightarrow \vec{k} = \vec{\lambda}\vec{i} + \vec{\lambda}\vec{j} + \vec{\lambda}\vec{k} + \vec{\mu}\vec{i} + \vec{\mu}\vec{j}$$

Equate corresponding coefficients on both sides

$$\vec{\lambda} + \vec{\mu} = 0 \text{ and } \vec{\lambda} = -1 \text{ this gives } \vec{\mu} = 1$$

$\therefore$ Then the value of $\vec{\lambda} + \vec{\mu} = 0$.

Question 9.

If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \vec{c} \sqrt{2}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(a) 81

(b) 9

(c) 27

(d) 18

Solution:

(a) 81

Hint:

$$\{[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}]\}_2 = \{[\vec{a} \times \vec{b}]\}_2 = \{(3)\}_2 = \{9\}_2 = 81$$

Question 10.

If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vector such that $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \vec{c} \sqrt{2}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(a) $\pi/2$

(b) $3\pi/4$

(c) $\pi/4$

(d) π

Solution:

(b) $3\pi/4$

Hint:

$$\vec{a} \times (\vec{b} \times \vec{c}) \equiv \vec{b} + \vec{c} \sqrt{2} \Rightarrow (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \equiv \vec{b} + \vec{c} \sqrt{2} \Rightarrow \vec{a} \cdot \vec{c} = 1, \vec{a} \cdot \vec{b} = -\sqrt{2} \Rightarrow \vec{a} \cdot \vec{b} = -1 \sqrt{2} \Rightarrow |\vec{a}| |\vec{b}| \cos \theta = -1 \sqrt{2} \Rightarrow \cos \theta = -1 \sqrt{2} \Rightarrow \theta = \cos^{-1}(-1 \sqrt{2}) = 3\pi/4$$

Question 11.

If the volume of the parallelepiped with $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ as coterminal edges is 8 cubic units, then the volume of the parallelepiped with $(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}), (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})$ and $(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})$ as coterminal edges is

(a) 8 cubic units

(b) 512 cubic units

(c) 64 cubic units

(d) 24 cubic units

Solution:

(c) 64 cubic units

Hint:

$$[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] = 8 \text{ (Given)} \Rightarrow [(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}), (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}), (\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})] = [(\vec{a} \times \vec{b}), (\vec{b} \times \vec{c}), (\vec{c} \times \vec{a})]_2 = (8)_2 = 64$$

Question 12.

Consider the vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 0$. Let P_1 and P_2 be the planes determined by the pairs of vectors $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ respectively. Then the angle between P_1 and P_2 is

(a) 0°

(b) 45°

(c) 60°

(d) 90°

Solution:

(a) 0°

Hint:

$$P_1 \rightarrow \vec{a}, \vec{b} \Rightarrow \vec{n}_1 \rightarrow \vec{a} \times \vec{b} \Rightarrow \vec{n}_1 \perp \vec{a}, \vec{b} \text{ Given } \vec{n}_1 \times \vec{n}_2 = 0 \Rightarrow \vec{n}_1, \vec{n}_2 \text{ are normal unit vectors } \Rightarrow |\vec{n}_1| |\vec{n}_2|$$

$$\sin \theta = 0 \Rightarrow \theta = 0^\circ$$

Question 13.

If $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$, where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors such that $\vec{b} \cdot \vec{c} \neq 0$ and $\vec{a} \cdot \vec{b} \neq 0$, then $\vec{a}$ and $\vec{c}$ are

(a) perpendicular

(b) parallel

(c) inclined at an angle $\pi/3$

(d) inclined at an angle $\pi/6$

Solution:

(b) parallel

Hint:

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{b} \cdot \vec{c})\vec{c} = \vec{\lambda} \vec{c}$$

$\therefore \vec{a}$ and $\vec{c}$ are parallel.

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Question 14.

If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{c} = 3\hat{i} + 5\hat{j} - \hat{k}$, then a vector perpendicular to $\vec{a}$ and lies in the plane containing $\vec{b}$ and $\vec{c}$ is

(a) $-17\hat{i} + 21\hat{j} - 97\hat{k}$

(b) $-17\hat{i} + 21\hat{j} - 123\hat{k}$

(c) $-17\hat{i} - 21\hat{j} + 97\hat{k}$

(d) $-17\hat{i} - 21\hat{j} - 97\hat{k}$

Solution:

(d) $-17\hat{i} - 21\hat{j} - 97\hat{k}$

Hint:

A vector $\perp$ to $\vec{a}$ and lies in the plane containing $\vec{b}$ and $\vec{c}$ is $\vec{a} \times (\vec{b} \times \vec{c})$

$$\vec{b} \times \vec{c} =$$

|

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -5 \\ 3 & 5 & -1 \end{vmatrix}$$

|

$$= \begin{vmatrix} 23 & -14 \\ -1 & 16 \end{vmatrix} = \vec{a} \times (\vec{b} \times \vec{c}) = 34 - 14 \times 1$$

|

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 23 & -123 & -14 \\ -1 & 16 & 1 \end{vmatrix}$$

|

$$= -17\hat{i} - 21\hat{j} - 97\hat{k}$$

Question 15.

The angle lines $x-23=y+1-2, z=2$ and $x-11=2y+33=z+52$ is

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/3$

(d) $\pi/2$

Solution:

(d) $\pi/2$

Hint: $x-23=y+1-2, z=2$ and $x-11=2y+33=z+52$ $x-11=y+33/2=z+52$

d.r.s : $(3, -2, 0)$ and d.c.s $(1, 32, 2)$

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} = \frac{3 \cdot 1 + (-2) \cdot 32 + 0 \cdot 2}{\sqrt{3^2 + (-2)^2 + 0^2} \sqrt{1^2 + 32^2 + 2^2}} = \frac{0}{\sqrt{13} \sqrt{1057}} = 0 \Rightarrow \theta = \pi/2$$

Question 16.

If the line $x-23=y-1-5=z+22$

lies in the plane $x+3\alpha z+\beta=0$, then (α, β) is

(a) $(-5, 5)$

(b) $(-6, 7)$

(c) $(5, -5)$

(d) $(6, -7)$

Solution:

(d) $(-6, 7)$

Hint:

d.c.s of the first line $= (3, -5, 2)$

d.c.s of the line perpendicular to plane $= (1, 3, -\alpha)$

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$3-15-2\alpha=0 \Rightarrow -12-2\alpha=0$$

$$-2\alpha=12 \Rightarrow \alpha=-6$$

Plane passes through the point (2,1,-2) so

$$2+3+6(-2)+\beta=0 \Rightarrow \beta=7$$

$$(\alpha, \beta)=(-6, 7)$$

Question 17.

The angle between the line $\vec{r}=(\hat{i}+2\hat{j}-3\hat{k})+t(2\hat{i}+\hat{j}-2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i}+\hat{j})+4=0$ is

(a) 0°

(b) 30°

(c) 45°

(d) 90°

Solution:

(c) 45°

Hint:

$$\vec{b}=2\hat{i}+\hat{j}-2\hat{k}; \vec{n}=\hat{i}+\hat{j} \sin \theta = \vec{b} \cdot \vec{n} / |\vec{b}| |\vec{n}| \sin \theta = 2+1 \sqrt{4+1+4} \sqrt{1+1} = 3 \sqrt{2} = 1 \sqrt{2} \Rightarrow \theta = 45^\circ$$

Question 18.

The coordinates of the point where the line $\vec{r}=(6\hat{i}-\hat{j}-3\hat{k})+t(-\hat{i}+4\hat{k})$ meets the plane $\vec{r} \cdot (\hat{i}+\hat{j}-\hat{k})$ are

(a) (2,1,0)

(b) (7,-1,-7)

(c) (1,2,-6)

(d) (5,-1,1)

Solution:

(d) (5,-1,1)

Hint:

Cartesian equation of the line

$$x-6-1=y+10+z+34=\lambda(-\lambda+6,-1,4\lambda-3)$$

This meets the plane $x+y-z=3$

$$-\lambda+6-1-4\lambda+3=3 \Rightarrow -5\lambda=-5$$

$$\lambda=1$$

The required point (5,-1,1).

Question 19.

Distance from the origin to the plane $3x-6y+2z+7=0$ is

(a) 0

(b) 1

(c) 2

(d) 3

Solution:

(b) 1

Hint:

Distance from the origin (0,0,0) to the plane

$$= \left| \frac{3(0)-6(0)+2(0)+7}{\sqrt{9+36+4}} \right| = \frac{7}{\sqrt{49}} = 1$$

Question 20.

The distance between the planes $x+2y+3z+7=0$ and $2x+4y+6z+7=0$ is

(a) $\frac{7\sqrt{2}}{2}$

(b) 72

(c) $\frac{7}{2}$

(d) $72\sqrt{2}$

Solution:

(a) $\frac{7\sqrt{2}}{2}$

Hint:

$$x+2y+3z+1=0; 2x+4y+6z+7=0$$

Multiplying 2 on both sides

$$2x+4y+6z+14=0, a=2, b=4, c=6, d_1=14, d_2=7$$

$$\text{Distance} = \sqrt{d_1^2 + d_2^2 + d_3^2} = \sqrt{14^2 + 7^2 + 7^2} = \sqrt{196 + 49 + 49} = \sqrt{294} = 7\sqrt{6}$$

Question 21.

If the direction cosines of a line are l, m, n , then

- (a) $l = \pm 3$
 (b) $l = \pm \sqrt{3}$
 (c) $l > 0$
 (d) $0 < l < 1$

Solution:

(b) $l = \pm \sqrt{3}$

Hint:

We know that sum of the squares of direction cosines = 1

$$l^2 + m^2 + n^2 = 1 \Rightarrow 3l^2 = 1 \Rightarrow l = \pm \sqrt{3}$$

Question 22.

The vector equation $\vec{r} = (\hat{i} - 2\hat{j} - \hat{k}) + t(\hat{j} - \hat{k})$ represents a straight line passing through the points and (0, -6, 1)

- (a) (0, 6, -1) and (1, -2, -1) (b) (0, 6, -1) and (-1, -4, -2) (c) (1, -2, -1) and (1, 4, -2) (d) (1, -2, -1)

Solution:

- (c) (1, -2, -1) and (1, 4, -2)

Hint:

The required vector equation is $\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$

$$\vec{r} = \hat{i} - 2\hat{j} - \hat{k} + t(\hat{j} - \hat{k}) = \hat{i} - 2\hat{j} - \hat{k} + t(\hat{j} - \hat{k}) = \hat{i} - 2\hat{j} - \hat{k} + t\hat{j} - t\hat{k} = \hat{i} + (t-2)\hat{j} - (1+t)\hat{k}$$

From (1) and (2) The points are (1, -2, -1) and (1, 4, -2)

Question 23.

If the distance of the point (1, 1, 1) from the origin is half of its distance from the plane $x + y + z + k = 0$, then the values of k are

- (a) ± 3 (b) ± 6 (c) -3, 9 (d) 3, -9

Solution:

- (d) 3, -9

Hint:

$$\sqrt{3} = \frac{1}{2} \frac{|1+1+1+k|}{\sqrt{1^2+1^2+1^2}} \Rightarrow \sqrt{3} = \frac{|3+k|}{2\sqrt{3}} \Rightarrow 3+k = \pm 6 \Rightarrow k = 3, -9$$

Question 24.

If the planes $\vec{r} \cdot (2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ are parallel, then the value of λ and μ are

Solution:

- (c) -12, -2

Hint:

$$\vec{a} = 2\hat{i} - \lambda\hat{j} + \hat{k} \text{ and } \vec{b} = 4\hat{i} + \hat{j} - \mu\hat{k}$$

Given $\vec{a}$ and $\vec{b}$ are parallel

$$\vec{a} = m\vec{b} \Rightarrow 2\hat{i} - \lambda\hat{j} + \hat{k} = m(4\hat{i} + \hat{j} - \mu\hat{k}) \Rightarrow 2 = 4m, -\lambda = m, 1 = -m\mu \Rightarrow m = \frac{1}{2}, \lambda = -\frac{1}{2}, \mu = -2$$

so

$$(\lambda, \mu) = (-1/2, -2)$$

Question 25.

If the length of the perpendicular from the origin to the plane $2x + 3y + \lambda z = 1, \lambda > 0$ is $\frac{1}{\sqrt{13}}$, then the value of λ is

- (a) $2\sqrt{3}$
 (b) $3\sqrt{2}$
 (c) 0
 (d) 1

Solution:

(a) $2\sqrt{3}$

Hint:

Given length of perpendicular from origin to the plane = $\frac{1}{\sqrt{13}}$

$$\frac{|2(0) + 3(0) + \lambda(0) - 1|}{\sqrt{2^2 + 3^2 + \lambda^2}} = \frac{1}{\sqrt{13}} \Rightarrow \frac{1}{\sqrt{13 + \lambda^2}} = \frac{1}{\sqrt{13}} \Rightarrow \lambda = 0$$

Squaring on both sides

$$25 = 13 + \lambda^2 \Rightarrow \lambda^2 = 12 \Rightarrow \lambda = \sqrt{12} = 2\sqrt{3}$$

Ex 6.10

Choose the correct or the most suitable answer from the given four alternatives:

Question 1.

If $\vec{a}$ and $\vec{b}$ are parallel vector, then $[\vec{a}, \vec{c}, \vec{b}]$ is equal to

- (a) 2
- (b) -1
- (c) 1
- (d) 0

Solution:

- (d) 0

Hint:

$\vec{a}$ and $\vec{b}$ are parallel vectors, so $\vec{a} \times \vec{b} = 0$

then $[\vec{a}, \vec{c}, \vec{b}] = -[\vec{a}, \vec{b}, \vec{c}] = -(\vec{a} \times \vec{b}) \cdot \vec{c} = 0$

Question 2.

If a vector $\vec{\alpha}$ lies in the plane of $\vec{\beta}$ and $\vec{\gamma}$, then .

- (a) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 1$
- (b) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = -1$
- (c) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$
- (d) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 2$

Solution:

- (c) $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$

Hint:

Since $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}]$ are lie in the same plane

so $[\vec{\alpha}, \vec{\beta}, \vec{\gamma}] = 0$

Question 3.

If $\vec{a} \perp \vec{b} \Rightarrow \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$, then the value of $[(\vec{a}, \vec{b}, \vec{c})]$ is

- (a) $|\vec{a}| |\vec{b}| |\vec{c}|$
- (b) $\frac{1}{3} |\vec{a}| |\vec{b}| |\vec{c}|$
- (c) 1
- (d) -1

Solution:

- (a) $|\vec{a}| |\vec{b}| |\vec{c}|$

Hint:

$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$

$\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors

$[(\vec{a}, \vec{b}, \vec{c})] = |\vec{a}| |\vec{b}| |\vec{c}|$ is volume of cuboid where $\vec{a}, \vec{b}, \vec{c}$ are coterminus edges.

Question 4.

If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}$ is perpendicular to $\vec{b}$, and is parallel to $\vec{c}$ then $\vec{a} \times (\vec{b} \times \vec{c})$ is equal to .

- (a) $\vec{a}$
- (b) $\vec{b}$
- (c) $\vec{c}$
- (d) $\vec{0}$

Solution:

- (b) $\vec{b}$

Hint:

$\vec{a}$ is perpendicular to $\vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$

$\vec{a}$ is parallel to $\vec{c} \Rightarrow \vec{a} \times \vec{c} = 0$

then $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

$\Rightarrow 0\vec{b} - \vec{0}\vec{c} = \vec{b}$

Question 5.

If $[\vec{a}, \vec{b}, \vec{c}] = 1$, then the value of $-(\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) - \vec{b} \cdot (\vec{c} \times \vec{a}) + (\vec{a} \times \vec{b}) \cdot \vec{c} + (\vec{a} \times \vec{b}) \cdot \vec{c}$ is

- (a) 1

(b) -1

(c) 2

(d) 3

Solution:

(a) 1

Hint:

$$[\vec{a}, \vec{b}, \vec{c}] = 1 \quad \text{(Given)} \quad \vec{a} \times (\vec{b} \times \vec{c}) + (\vec{c} \times \vec{a}) \times \vec{b} + \vec{b} \times (\vec{c} \times \vec{a}) = \vec{a} + \vec{b} + \vec{c} \quad \text{--- (1)}$$
$$[\vec{a}, \vec{b}, \vec{c}] = 1 \quad \text{--- (2)}$$

Question 6.

The volume of the parallelepiped with its edges represented by the vectors $\hat{i} + \hat{j}$, $\hat{i} + 2\hat{j}$, $\hat{i} + \hat{j} + \pi\hat{k}$ is

(a) π^2

(b) π^3

(c) π

(d) π^4

Solution:

(c) π

Hint:

$$\text{Volume} = |[\vec{a}, \vec{b}, \vec{c}]|$$

$$= |110 \ 120 \ 11\pi|$$

$$= \pi(2-1) = \pi \text{ cubic units}$$

Question 7.

If $\vec{a}$ and $\vec{b}$ are unit vectors such that $[\vec{a}, \vec{b}, \vec{a} \times \vec{b}] = \pi/4$ then the angle between $\vec{a}$ and $\vec{b}$ is

(a) $\pi/6$

(b) $\pi/4$

(c) $\pi/3$

(d) $\pi/2$

Solution:

(a) $\pi/6$

Hint:

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$[\vec{a}, \vec{b}, \vec{a} \times \vec{b}] = \frac{1}{4}$		$ \vec{a} \vec{b} \sin \theta = \frac{1}{2}$
$(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) = \frac{1}{4}$		$(1)(1) \sin \theta = \frac{1}{2}$
$(\vec{a} \times \vec{b})^2 = \frac{1}{4}$		$\sin \theta = \frac{1}{2}$
$ \vec{a} \times \vec{b} ^2 = \frac{1}{4}$		$\theta = \sin^{-1}\left(\frac{1}{2}\right)$
$ \vec{a} \times \vec{b} = \frac{1}{2}$		$\theta = \frac{\pi}{6}$

Question 8.

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j}$, $\vec{c} = \hat{i}$ and $(\vec{a} \times \vec{b}) \times \vec{c} = \lambda \vec{a} + \mu \vec{b}$, then the value of $\lambda + \mu$ is

(a) 0

(b) 1

(c) 6

(d) 3

Solution:

(a) 0

Hint:

$$\text{But } (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} \equiv \vec{b} - \vec{a} = \vec{a} \times \vec{b} \times \vec{c} = \vec{\lambda}\vec{a} + \vec{\mu}\vec{b} \Rightarrow \vec{a} = \vec{\lambda}\vec{a} + \vec{\mu}\vec{b} \Rightarrow \vec{k} = \vec{\lambda}\vec{i} + \vec{\lambda}\vec{j} + \vec{\lambda}\vec{k} + \vec{\mu}\vec{i} + \vec{\mu}\vec{j}$$

Equate corresponding coefficients on both sides

$$\vec{\lambda} + \vec{\mu} = 0 \text{ and } \vec{\lambda} = -1 \text{ this gives } \vec{\mu} = 1$$

$\therefore$ Then the value of $\vec{\lambda} + \vec{\mu} = 0$.

Question 9.

If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \sqrt{2}$, then the angle between $\vec{a}$ and $\vec{b}$ is

- (a) 81
- (b) 9
- (c) 27
- (d) 18

Solution:

- (a) 81

Hint:

$$[(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{a})]_2 = [(\vec{a} \times \vec{b})]_2 = \{(3)_2\}_2 = \{9\}_2 = 81$$

Question 10.

If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vector such that $\vec{a} \times (\vec{b} \times \vec{c}) = \sqrt{2}$, then the angle between $\vec{a}$ and $\vec{b}$ is

- (a) $\pi/2$
- (b) $3\pi/4$
- (c) $\pi/4$
- (d) π

Solution:

- (b) $3\pi/4$

Hint:

$$\vec{a} \times (\vec{b} \times \vec{c}) \equiv \sqrt{2} \Rightarrow (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \sqrt{2} \Rightarrow \vec{a} \cdot \vec{c} = 1, \vec{a} \cdot \vec{b} = -1 \Rightarrow \cos \theta = -1/\sqrt{2} \Rightarrow \theta = \cos^{-1}(-1/\sqrt{2}) = 3\pi/4$$

Question 11.

If the volume of the parallelepiped with $\vec{a}, \vec{b}, \vec{c}$ as coterminous edges is 8 cubic units, then the volume of the parallelepiped with $(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}), (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})$ and $(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})$ as coterminous edges is

- (a) 8 cubic units
- (b) 512 cubic units
- (c) 64 cubic units
- (d) 24 cubic units

Solution:

- (c) 64 cubic units

Hint:

$$[(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}), (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}), (\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})] = [(\vec{a} \times \vec{b}), (\vec{b} \times \vec{c}), (\vec{c} \times \vec{a})]_2 = (8)_2 = 64$$

Question 12.

Consider the vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 0$. Let P_1 and P_2 be the planes determined by the pairs of vectors $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ respectively. Then the angle between P_1 and P_2 is

- (a) 0°
- (b) 45°
- (c) 60°
- (d) 90°

Solution:

- (a) 0°

Hint:

$$P_1 \rightarrow \vec{a}, \vec{b} \Rightarrow \vec{n}_1 = \vec{a} \times \vec{b} \Rightarrow \vec{n}_2 = \vec{c} \times \vec{d} \text{ Given: } \vec{n}_1 \times \vec{n}_2 = 0 \Rightarrow \vec{n}_1, \vec{n}_2 \text{ are normal unit vectors } \Rightarrow |\vec{n}_1| |\vec{n}_2| \sin \theta = 0 \Rightarrow \sin \theta = 0 \Rightarrow \theta = 0^\circ$$

Question 13.

If $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$, where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors such that $\vec{b} \times \vec{c} \neq 0$ and $\vec{a} \cdot \vec{b} \neq 0$, then $\vec{a}$ and $\vec{c}$ are

- (a) perpendicular
- (b) parallel
- (c) inclined at an angle $\pi/3$
- (d) inclined at an angle $\pi/6$

Solution:

(b) parallel

Hint:

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c} - (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} = (\vec{a} \cdot \vec{b}\vec{b} \cdot \vec{c} - \vec{c} \cdot \vec{a})\vec{c} = \vec{a} \cdot \vec{c}\vec{c}$$

$\therefore \vec{a}$ and $\vec{c}$ are parallel.

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Question 14.

If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{c} = 3\hat{i} + 5\hat{j} - \hat{k}$, then a vector perpendicular to $\vec{a}$ and lies in the plane containing $\vec{b}$ and $\vec{c}$ is

- (a) $-17\hat{i} + 21\hat{j} - 97\hat{k}$
- (b) $-17\hat{i} + 21\hat{j} - 123\hat{k}$
- (c) $-17\hat{i} - 21\hat{j} + 97\hat{k}$
- (d) $-17\hat{i} - 21\hat{j} - 97\hat{k}$

Solution:

- (d) $-17\hat{i} - 21\hat{j} - 97\hat{k}$

Hint:

A vector $\perp$ to $\vec{a}$ and lies in the plane containing $\vec{b}$ and $\vec{c}$ is $\vec{a} \times (\vec{b} \times \vec{c})$

$$\vec{b} \times \vec{c} =$$

|

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -5 \\ 3 & 5 & -1 \end{vmatrix}$$

|

$$= 23\hat{i} - 14\hat{j} - \hat{k} \quad \vec{a} \times (\vec{b} \times \vec{c}) =$$

|

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 23 & -14 & -1 \end{vmatrix}$$

|

$$= -17\hat{i} - 21\hat{j} - 97\hat{k}$$

Question 15.

The angle lines $x-23=y+1-2, z=2$ and $x-11=2y+33=z+52$ is

- (a) $\pi/6$
- (b) $\pi/4$
- (c) $\pi/3$
- (d) $\pi/2$

Solution:

- (d) $\pi/2$

Hint: $x-23=y+1-2, z=2$ and $x-11=2y+33=z+52$ $x-11=y+32, z=52$

d.r.s : $(3, -2, 0)$ and d.c.s $(1, 32, 2)$

$$\cos \theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} = \frac{3-6+0}{\sqrt{3^2 + (-2)^2 + 0^2} \sqrt{1^2 + 32^2 + 2^2}} = \frac{-3}{\sqrt{13} \sqrt{1057}} = 0 \Rightarrow \theta = \frac{\pi}{2}$$

Question 16.

If the line $x-23=y-1-5=z+22$

lies in the plane $x+3\alpha+z+\beta=0$, then (α, β) is

- (a) $(-5, 5)$
- (b) $(-6, 7)$
- (c) $(5, -5)$
- (d) $(6, -7)$

Solution:

- (d) $(-6, 7)$

Hint:

d.c.s of the first line $= (3, -5, 2)$

d.c.s of the line perpendicular to plane $= (1, 3, -\alpha)$

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$3-15-2\alpha=0 \Rightarrow -12-2\alpha=0$$

$$-2\alpha = 12 \Rightarrow \alpha = -6$$

Plane passes through the point (2,1,-2) so

$$2+3+6(-2)+\beta=0 \Rightarrow \beta=7$$

$$(\alpha, \beta) = (-6, 7)$$

Question 17.

The angle between the line $\vec{r} = (\hat{i} + 2\hat{j} - 3\hat{k}) + t(2\hat{i} + \hat{j} - 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} + \hat{j}) + 4 = 0$ is

(a) 0°

(b) 30°

(c) 45°

(d) 90°

Solution:

(c) 45°

Hint:

$$\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}; \vec{n} = \hat{i} + \hat{j} \sin \theta = \vec{b} \cdot \vec{n} / |\vec{b}| |\vec{n}| \sin \theta = 2 + 1 \sqrt{4+1+4} \sqrt{1+1} = 33 \sqrt{2} = 1 \sqrt{2} \theta = 45^\circ$$

Question 18.

The coordinates of the point where the line $\vec{r} = (6\hat{i} - \hat{j} - 3\hat{k}) + t(-\hat{i} + 4\hat{k})$ meets the plane $\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k})$ are

(a) (2,1,0)

(b) (7,-1,-7)

(c) (1,2,-6)

(d) (5,-1,1)

Solution:

(d) (5,-1,1)

Hint:

Cartesian equation of the line

$$x-6-1=y+10+z+34=\lambda(-\lambda+6,-1,4\lambda-3)$$

This meets the plane $x+y-z=3$

$$-\lambda+6-1-41+3=3 \Rightarrow -5\lambda=-5$$

$$\lambda=1$$

The required point (5,-1,1).

Question 19.

Distance from the origin to the plane $3x-6y+2z+7=0$ is

(a) 0

(b) 1

(c) 2

(d) 3

Solution:

(b) 1

Hint:

Distance from the origin (0,0,0) to the plane

$$= \left| \frac{3(0)-6(0)+2(0)+7}{\sqrt{9+36+4}} \right| = \frac{7}{\sqrt{49}} = 1$$

Question 20.

The distance between the planes $x+2y+3z+7=0$ and $2x+4y+6z+7=0$ is

(a) $\frac{7\sqrt{2}}{2}$

(b) 72

(c) $\frac{7}{2}$

(d) $72\sqrt{2}$

Solution:

(a) $\frac{7\sqrt{2}}{2}$

Hint:

$$x+2y+3z+1=0; 2x+4y+6z+7=0$$

Multiplying 2 on both sides

$$2x+4y+6z+14=0, a=2, b=4, c=6, d_1=14, d_2=?$$

$$\text{Distance} = \left| \frac{d_1 - d_2}{\sqrt{a^2 + b^2 + c^2}} \right| = \left| \frac{14 - 7}{\sqrt{4 + 16 + 36}} \right| = \left| \frac{7}{\sqrt{56}} \right| = \frac{7\sqrt{2}}{2\sqrt{2}\sqrt{7}} = \frac{7\sqrt{2}}{2}$$

Question 21.

If the direction cosines of a line are $1c, 1c, 1c$, then

- (a) $c = \pm 3$
- (b) $c = \pm \sqrt{3}$
- (c) $c > 0$
- (d) $0 < c < 1$

Solution:

(b) $c = \pm \sqrt{3}$

Hint:

We know that sum of the squares of direction cosines = 1

$$1c^2 + 1c^2 + 1c^2 = 1 \Rightarrow 3c^2 = 1 \Rightarrow c = \pm \sqrt{3}$$

Question 22.

The vector equation $\vec{r} = (\hat{i} - 2\hat{j} - \hat{k}) + t(\hat{j} - \hat{k})$ represents a straight line passing through the points and (0, -6, 1)

- (a) (0, 6, -1) and (1, -2, -1) (b) (0, 6, -1) and (-1, -4, -2) (c) (1, -2, -1) and (1, 4, -2) (d) (1, -2, -1)

Solution:

- (c) (1, -2, -1) and (1, 4, -2)

Hint:

The required vector equation is $\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$

$$\vec{r} = 2\hat{i} - 3\hat{j} + 6\hat{k} + t(6\hat{i} - 6\hat{j} - 6\hat{k}) = 2\hat{i} - 3\hat{j} + 6\hat{k} + 6t(\hat{i} - \hat{j} - \hat{k})$$

From (1) and (2) The points are (1, -2, -1) and (1, 4, -2)

Question 23.

If the distance of the point (1, 1, 1) from the origin is half of its distance from the plane $x + y + z + k = 0$, then the values of k are

- (a) ± 3 (b) ± 6 (c) -3, 9 (d) 3, -9

Solution:

- (d) 3, -9

Hint:

$$\sqrt{3} = \frac{1}{\sqrt{3}} \frac{|1 + 1 + 1 + k|}{\sqrt{1^2 + 1^2 + 1^2}} \Rightarrow \sqrt{3} = \frac{|3 + k|}{3} \Rightarrow 3 + k = \pm 3 \Rightarrow k = 0 \text{ or } k = -6$$

Question 24.

If the planes $\vec{r} \cdot (2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ are parallel, then the value of λ and μ are

Solution:

- (c) -12, -2

Hint:

$$\vec{a} = 2\hat{i} - \lambda\hat{j} + \hat{k} \text{ and } \vec{b} = 4\hat{i} + \hat{j} - \mu\hat{k}$$

Given $\vec{a}$ and $\vec{b}$ are parallel

$$\vec{a} = m\vec{b} \Rightarrow 2\hat{i} - \lambda\hat{j} + \hat{k} = m(4\hat{i} + \hat{j} - \mu\hat{k}) \Rightarrow 4m = 2 \Rightarrow m = 1/2 \Rightarrow -\lambda = m \Rightarrow \lambda = -1/2 \Rightarrow 1 = -m\mu \Rightarrow \mu = -2$$

so

$$(\lambda, \mu) = (-1/2, -2)$$

Question 25.

If the length of the perpendicular from the origin to the plane $2x + 3y + \lambda z = 1, \lambda > 0$ is $\sqrt{15}$, then the value of λ is

- (a) $2\sqrt{3}$
- (b) $3\sqrt{2}$
- (c) 0
- (d) 1

Solution:

(a) $2\sqrt{3}$

Hint:

Given length of perpendicular from origin to the plane = $\sqrt{15}$

$$\frac{|2(0) + 3(0) + \lambda(0) - 1|}{\sqrt{2^2 + 3^2 + \lambda^2}} = \sqrt{15} \Rightarrow \frac{1}{\sqrt{13 + \lambda^2}} = \sqrt{15} \Rightarrow \sqrt{13 + \lambda^2} = \frac{1}{\sqrt{15}} \Rightarrow 13 + \lambda^2 = \frac{1}{15} \Rightarrow \lambda^2 = \frac{1}{15} - 13 \Rightarrow \lambda^2 = -\frac{194}{15} \Rightarrow \lambda = \frac{\sqrt{194}}{\sqrt{15}}$$

Squaring on both sides

$$25 = 13 + \lambda^2 \Rightarrow \lambda^2 = 12 \Rightarrow \lambda = \sqrt{12} \Rightarrow \lambda = 2\sqrt{3}$$